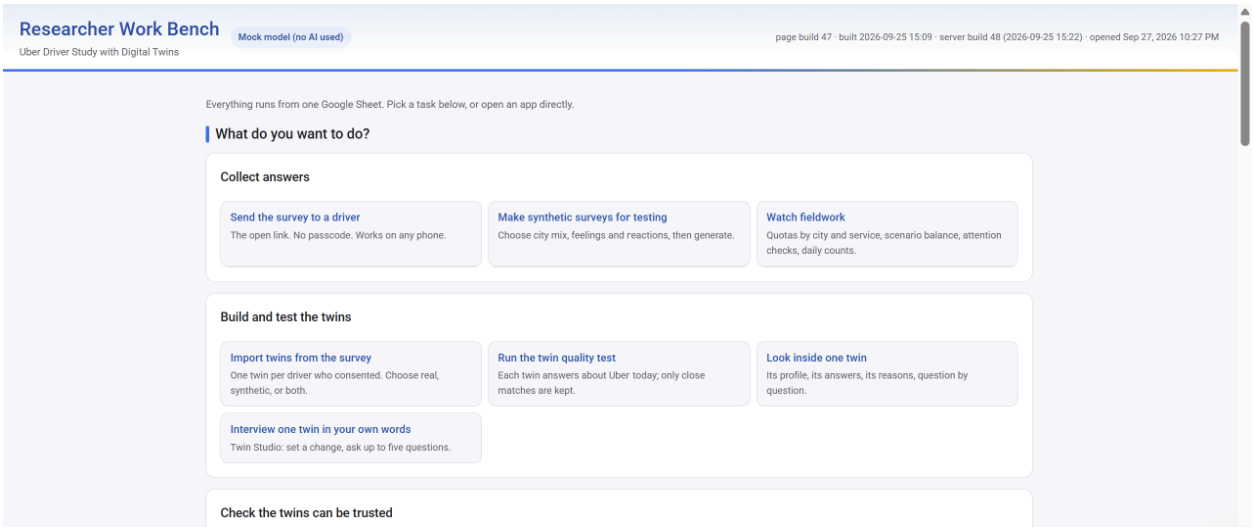
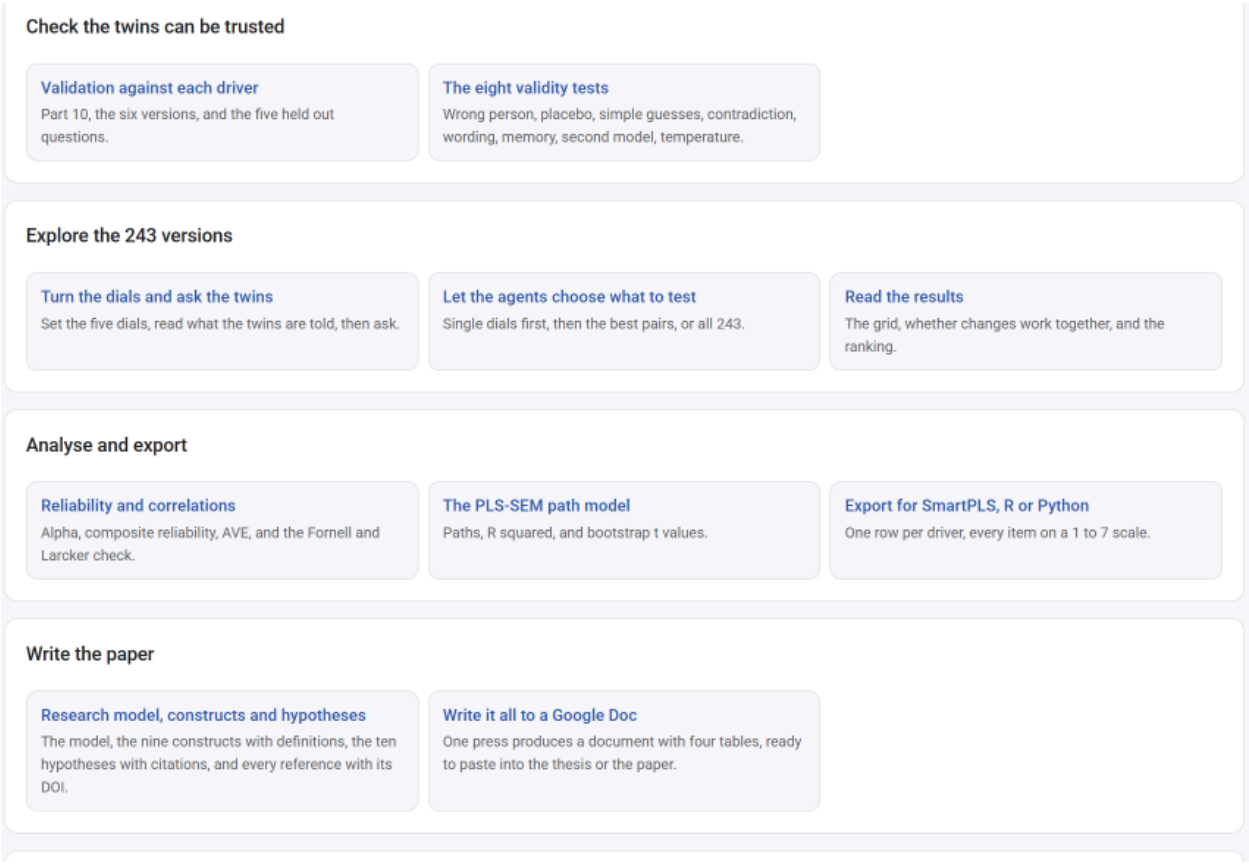
Main Menu


Researcher Work Bench
Mock model (no AI used)
Uber Driver Study with Digital Twins
page build 47 · built 2026-09-25 15:09 · server build 48 (2026-09-25 15:22) · opened Sep 27, 2026 10:27 PM
Everything runs from one Google Sheet. Pick a task below, or open an app directly.
What do you want to do?
Collect answers
Send the survey to a driver
The open link. No passcode. Works on any phone.
Make synthetic surveys for testing
Choose city mix, feelings and reactions, then generate.
Watch fieldwork
Quotas by city and service, scenario balance, attention checks, daily counts.
Build and test the twins
Import twins from the survey
One twin per driver who consented. Choose real, synthetic, or both.
Run the twin quality test
Each twin answers about Uber today; only close matches are kept.
Look inside one twin
Its profile, its answers, its reasons, question by question.
Interview one twin in your own words
Twin Studio: set a change, ask up to five questions.
Check the twins can be trusted


Check the twins can be trusted
Validation against each driver
Part 10, the six versions, and the five held out questions.
The eight validity tests
Wrong person, placebo, simple guesses, contradiction, wording, memory, second model, temperature.
Explore the 243 versions
Turn the dials and ask the twins
Set the five dials, read what the twins are told, then ask.
Let the agents choose what to test
Single dials first, then the best pairs, or all 243.
Read the results
The grid, whether changes work together, and the ranking.
Analyse and export
Reliability and correlations
Alpha, composite reliability, AVE, and the Fornell and Larcker check.
The PLS-SEM path model
Paths, R squared, and bootstrap t values.
Export for SmartPLS, R or Python
One row per driver, every item on a 1 to 7 scale.
Write the paper
Research model, constructs and hypotheses
The model, the nine constructs with definitions, the ten hypotheses with citations, and every reference with its DOI.
Write it all to a Google Doc
One press produces a document with four tables, ready to paste into the thesis or the paper.

Set up and manage

Model, speed and sample settings
Mock or live model, pause, calls at once, twins per run, token limit.

Agent instructions and the freeze
What each agent is told, its model, the rules, and the freeze before the main study.

Open the Google Sheet
Every tab of raw data.

Open an app directly

Driver survey

For drivers. No passcode.

The full survey: eligibility, consent, profile, 37 ratings with reasons, five open questions, five held out questions, one imagined change, six versions.

Open

Generator

For you.

Create test surveys from settings you choose, so you can try everything before real drivers arrive.

Open

Twin Lab

For you.

Import twins, run the quality test, validation and the eight validity tests, turn the dials, watch fieldwork, read results.

Open

Twin Studio

For you.

Build one twin and interview it in your own words. The place to check a prompt before a long run.

Open

Agent Studio

For you.

What each agent is told, which model it uses, the rules it follows, and the freeze before the main study.

Open

Statistics

For you.

Screening, reliability, correlations, a PLS-SEM path model, and an export for SmartPLS, R or Python.

Open

Research model

For you.

Constructs, hypotheses and references, and a button that writes them into a Google Doc.

Open

Google Sheet

The data.

Every tab: survey answers, twins, runs, designs, validation, validity tests, agents, and the log.

Open

Where things stand

Surveys saved	0 real, 34 synthetic
Twins built	34, 33 passed the quality test
Versions tested	7 of 243
Twin answers saved	8507
Validation run	yes, 59 twins checked
Validity tests run	10
Lab passcode	set

Quick start

1. **Make test data.** Open the *Generator*, choose a sentiment preset, set how many surveys, and press Generate.
2. **Build twins.** In *Twin Lab*, Settings tab, choose which answers to use, then Import twins from the survey.
3. **Test the twins.** Agents tab, then *1. Run twin quality test.*
4. **Check they predict.** Validation tab, then *Check twins against their drivers.*

Quick start

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4. **Check they predict.** Validation tab, then *Check twins against their drivers.*
5. **Ask awkward questions.** Validity tests tab: wrong person, beating simple guesses, and the rest.
6. **Turn the dials.** Dial board tab, set a version, then Ask the twins. Results tab shows the grid.
7. **Look at the data.** Statistics: reliability, correlations, PLS-SEM, and the export for SmartPLS or Python.

With the mock model this takes about fifteen minutes and costs nothing. Switch to a live AI model only when you are ready.

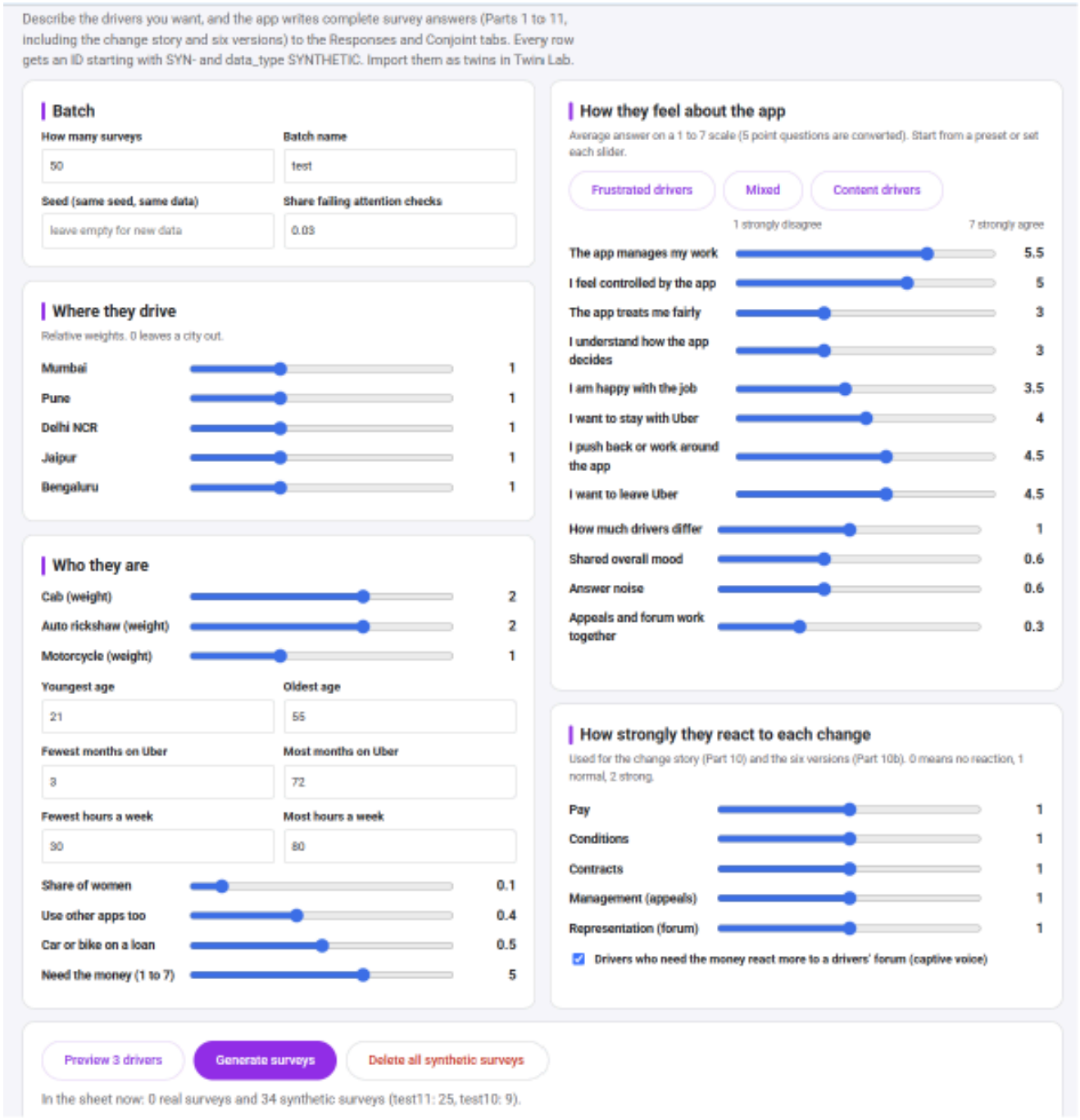
Help

- ▸ Turn on a live AI model
- ▸ A live run is slow, or the provider refuses calls
- ▸ A twin answer was refused
- ▸ Something is missing after an update
- ▸ Before real fieldwork

Main Navbar

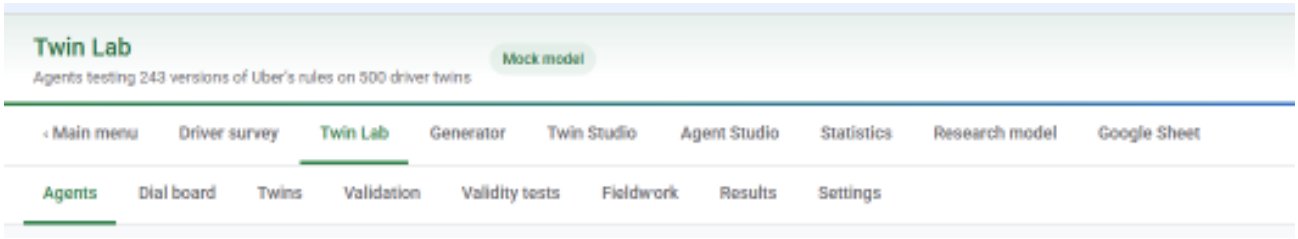


Synthetic Survey Generator

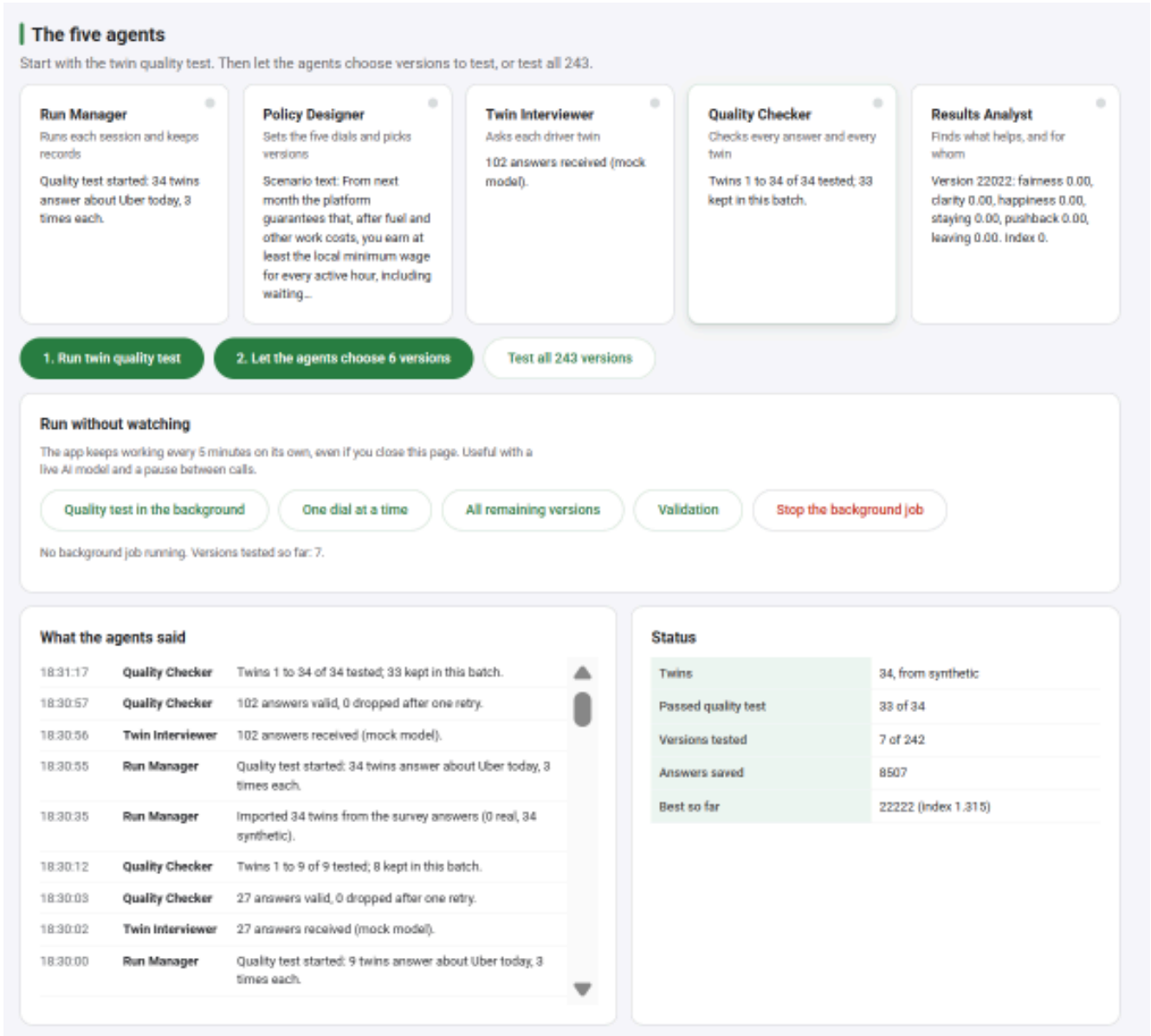


Twin Lab

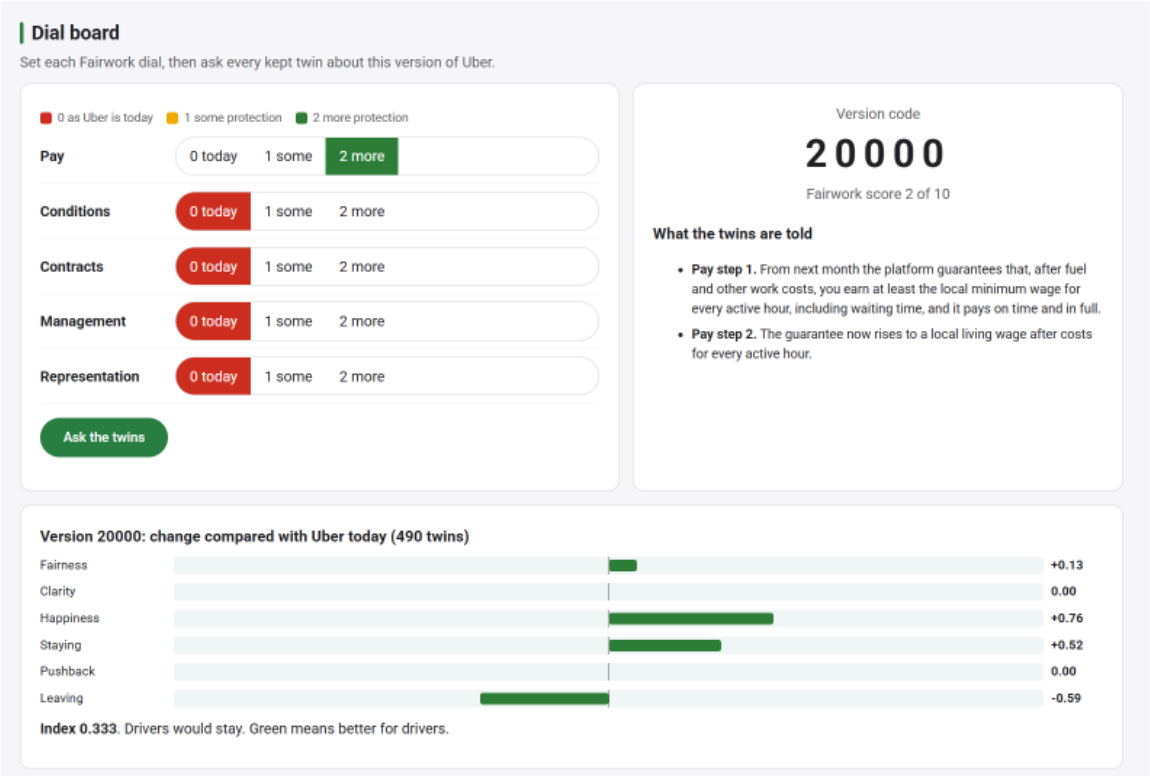
Main Navbar + Subnavbar



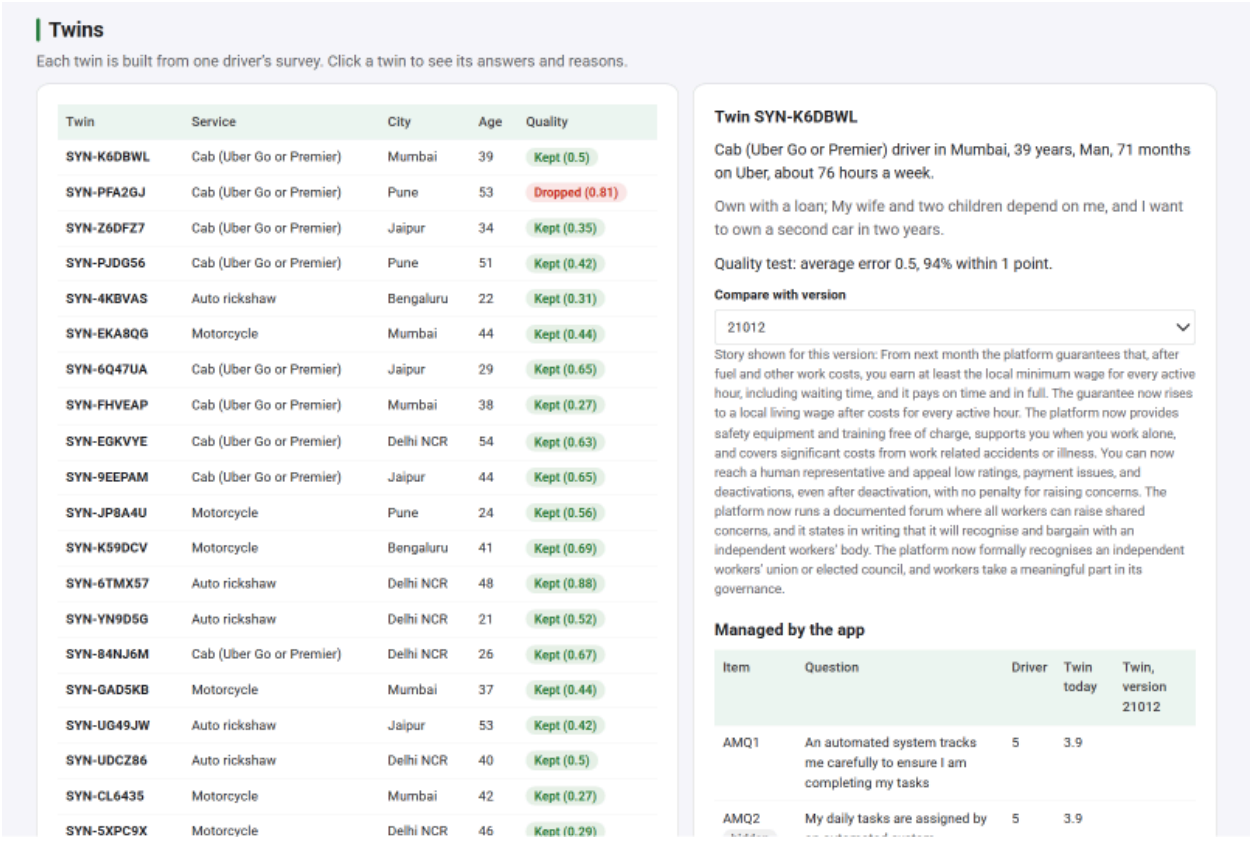
Twin Lab - > Agents Tab



Twin Lab -> Dial Board



Twin Lab - > Twin Tab



Twin	Service	City	Age	Quality
SYN-K6DBWL	Cab (Uber Go or Premier)	Mumbai	39	Kept (0.5)
SYN-PFA2GJ	Cab (Uber Go or Premier)	Pune	53	Dropped (0.81)
SYN-Z6DFZ7	Cab (Uber Go or Premier)	Jaipur	34	Kept (0.35)
SYN-PJDG56	Cab (Uber Go or Premier)	Pune	51	Kept (0.42)
SYN-4KBVAS	Auto rickshaw	Bengaluru	22	Kept (0.31)
SYN-EKA8QG	Motorcycle	Mumbai	44	Kept (0.44)
SYN-6Q47UA	Cab (Uber Go or Premier)	Jaipur	29	Kept (0.65)
SYN-FHVEAP	Cab (Uber Go or Premier)	Mumbai	38	Kept (0.27)
SYN-EGKVYE	Cab (Uber Go or Premier)	Delhi NCR	54	Kept (0.63)
SYN-9EEPAM	Cab (Uber Go or Premier)	Jaipur	44	Kept (0.65)
SYN-JP8A4U	Motorcycle	Pune	24	Kept (0.56)
SYN-K59DCV	Motorcycle	Bengaluru	41	Kept (0.69)
SYN-6TMX57	Auto rickshaw	Delhi NCR	48	Kept (0.88)
SYN-YN9D5G	Auto rickshaw	Delhi NCR	21	Kept (0.52)
SYN-84NJ6M	Cab (Uber Go or Premier)	Delhi NCR	26	Kept (0.67)
SYN-GAD5KB	Motorcycle	Mumbai	37	Kept (0.44)
SYN-UG49JW	Auto rickshaw	Jaipur	53	Kept (0.42)
SYN-UDCZ86	Auto rickshaw	Delhi NCR	40	Kept (0.5)
SYN-CL6435	Motorcycle	Mumbai	42	Kept (0.27)
SYN-5XPC9X	Motorcycle	Delhi NCR	46	Kept (0.29)

Twin Lab -> Twin Tab - > Answer of each twin

Managed by the app

Item	Question	Driver	Twin today	Twin, version 21012
AMQ1	An automated system tracks me carefully to ensure I am completing my tasks	5	3.9	
AMQ2 hidden	My daily tasks are assigned by an automated system	5	3.9	
AMQ3	My work schedule is made by an automated system	5	3.9	
AMQ4 hidden	The evaluation of my work performance is handled by an electronic system	5	4.5	
AMQ5	A large part of my compensation is determined by an automated system	6	4.7	

Twin's reason

Today: As a cab driver in Mumbai, the app decides almost everything about my day.

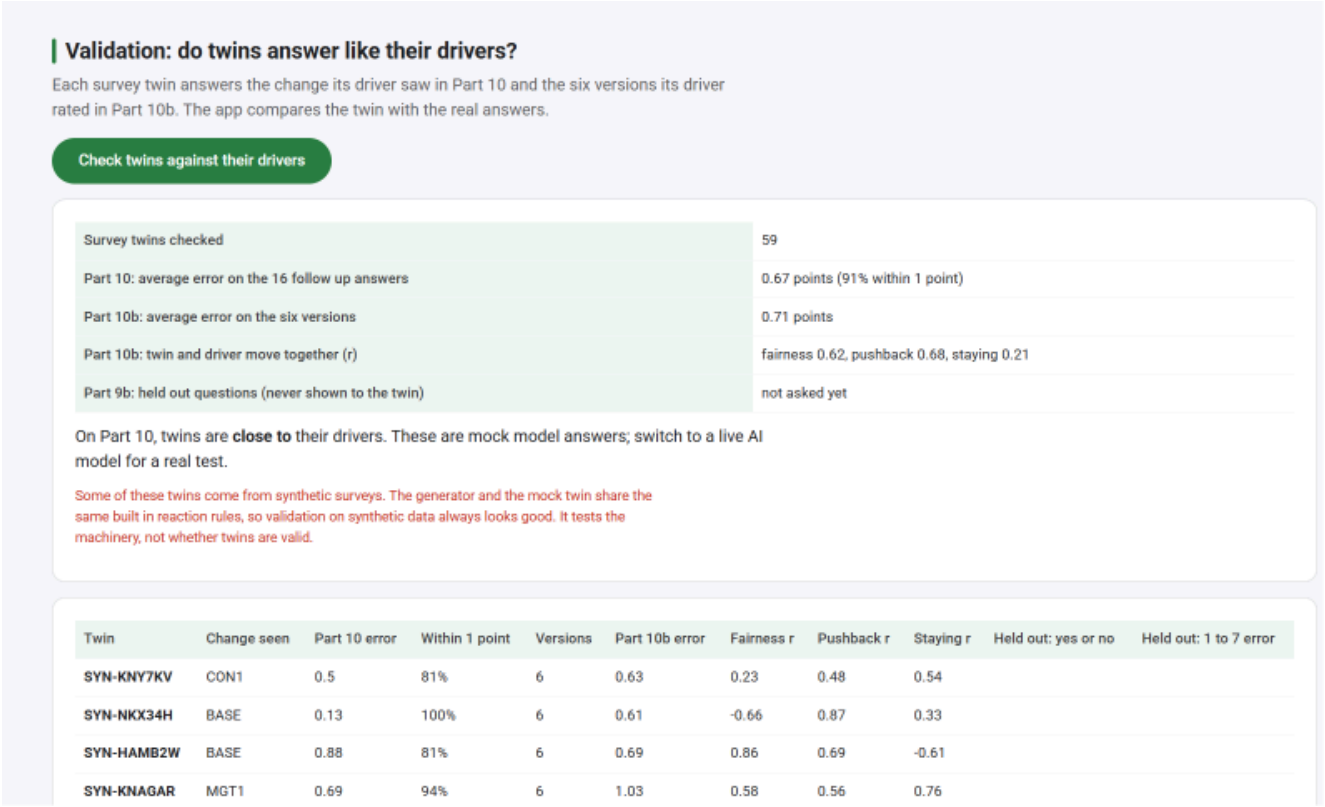
Version 21012: Not tested yet

Feels controlled

Item	Question	Driver	Twin today	Twin, version 21012
PAC1	The algorithm assigns my tasks intelligently	3	2.3	
PAC2 hidden	The algorithm provides normative instructions for my	3	2.3	

Twin Lab -> Validation Tab

Checking validation with Held out responses



Validation: do twins answer like their drivers?

Each survey twin answers the change its driver saw in Part 10 and the six versions its driver rated in Part 10b. The app compares the twin with the real answers.

Check twins against their drivers

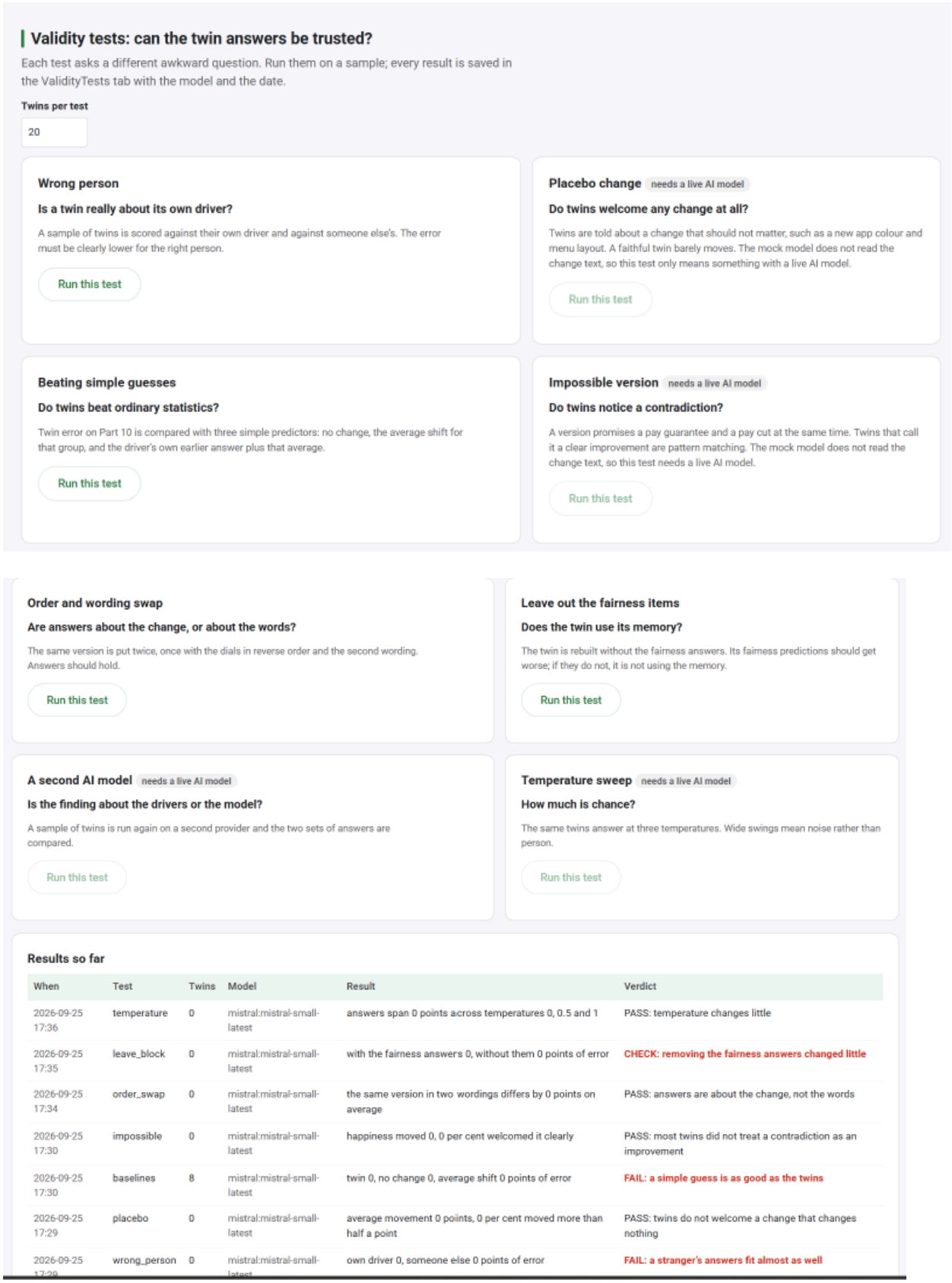
Survey twins checked	59
Part 10: average error on the 16 follow up answers	0.67 points (91% within 1 point)
Part 10b: average error on the six versions	0.71 points
Part 10b: twin and driver move together (r)	fairness 0.62, pushback 0.68, staying 0.21
Part 9b: held out questions (never shown to the twin)	not asked yet

On Part 10, twins are **close to** their drivers. These are mock model answers; switch to a live AI model for a real test.

Some of these twins come from synthetic surveys. The generator and the mock twin share the same built in reaction rules, so validation on synthetic data always looks good. It tests the machinery, not whether twins are valid.

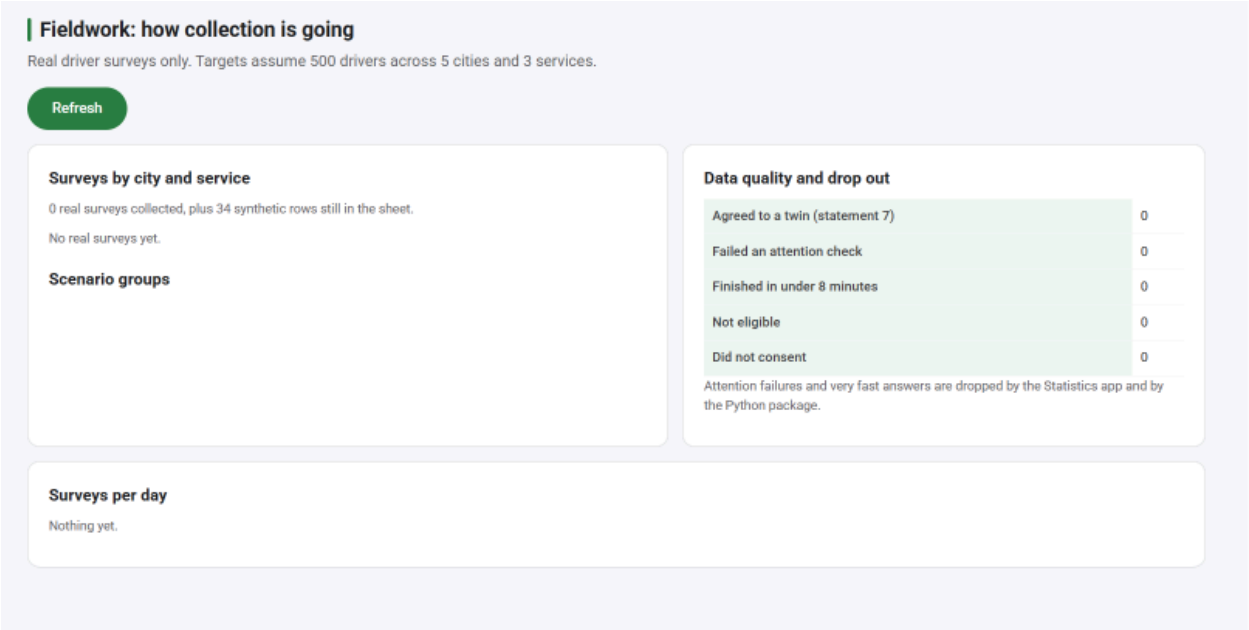
Twin	Change seen	Part 10 error	Within 1 point	Versions	Part 10b error	Fairness r	Pushback r	Staying r	Held out: yes or no	Held out: 1 to 7 error
SYN-KNY7KV	CON1	0.5	81%	6	0.63	0.23	0.48	0.54		
SYN-NKX34H	BASE	0.13	100%	6	0.61	-0.66	0.87	0.33		
SYN-HAMB2W	BASE	0.88	81%	6	0.69	0.86	0.69	-0.61		
SYN-KNAGAR	MGT1	0.69	94%	6	1.03	0.58	0.56	0.76		

Twin Lab -> Validity Test

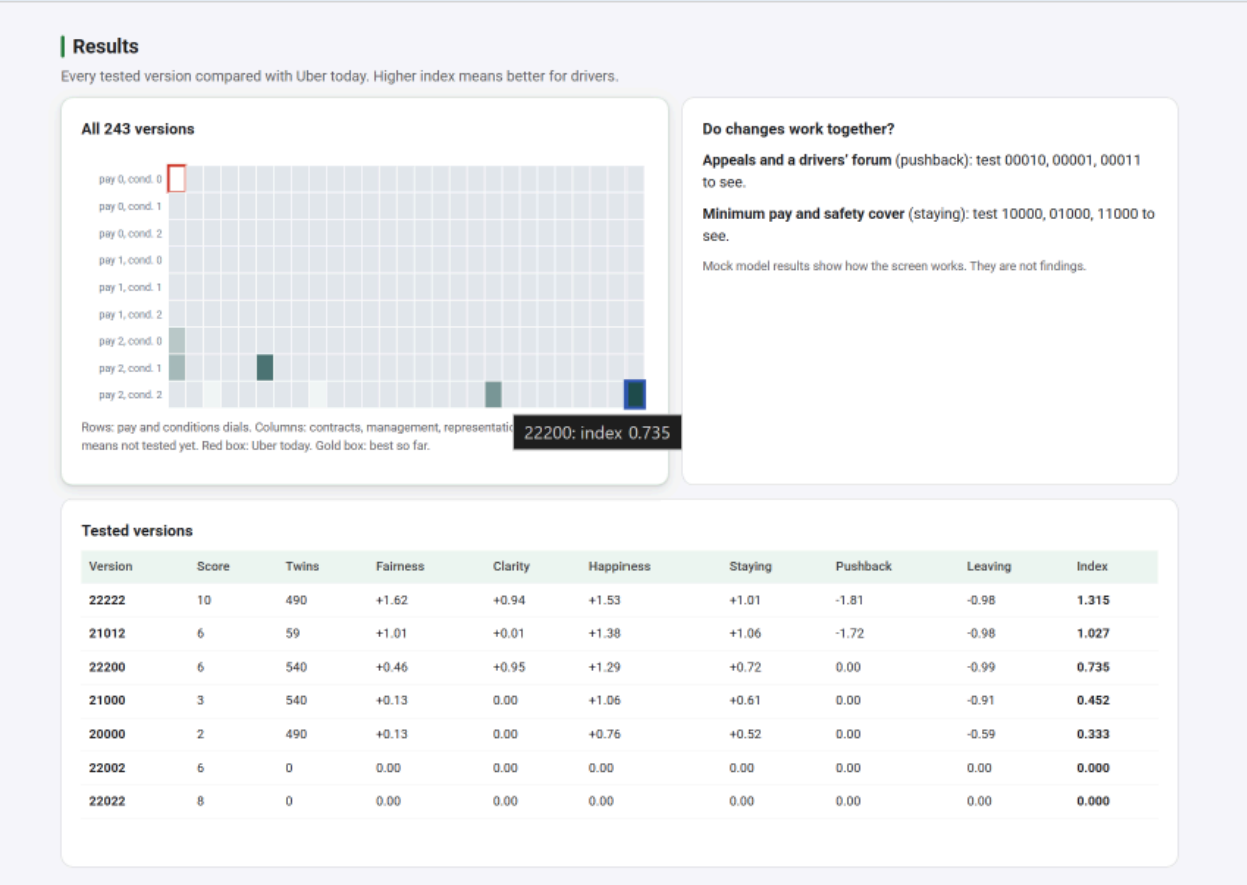


Build the backend of the tests as shown what to do, and also results so far are also the logs as and when each test is done, appended

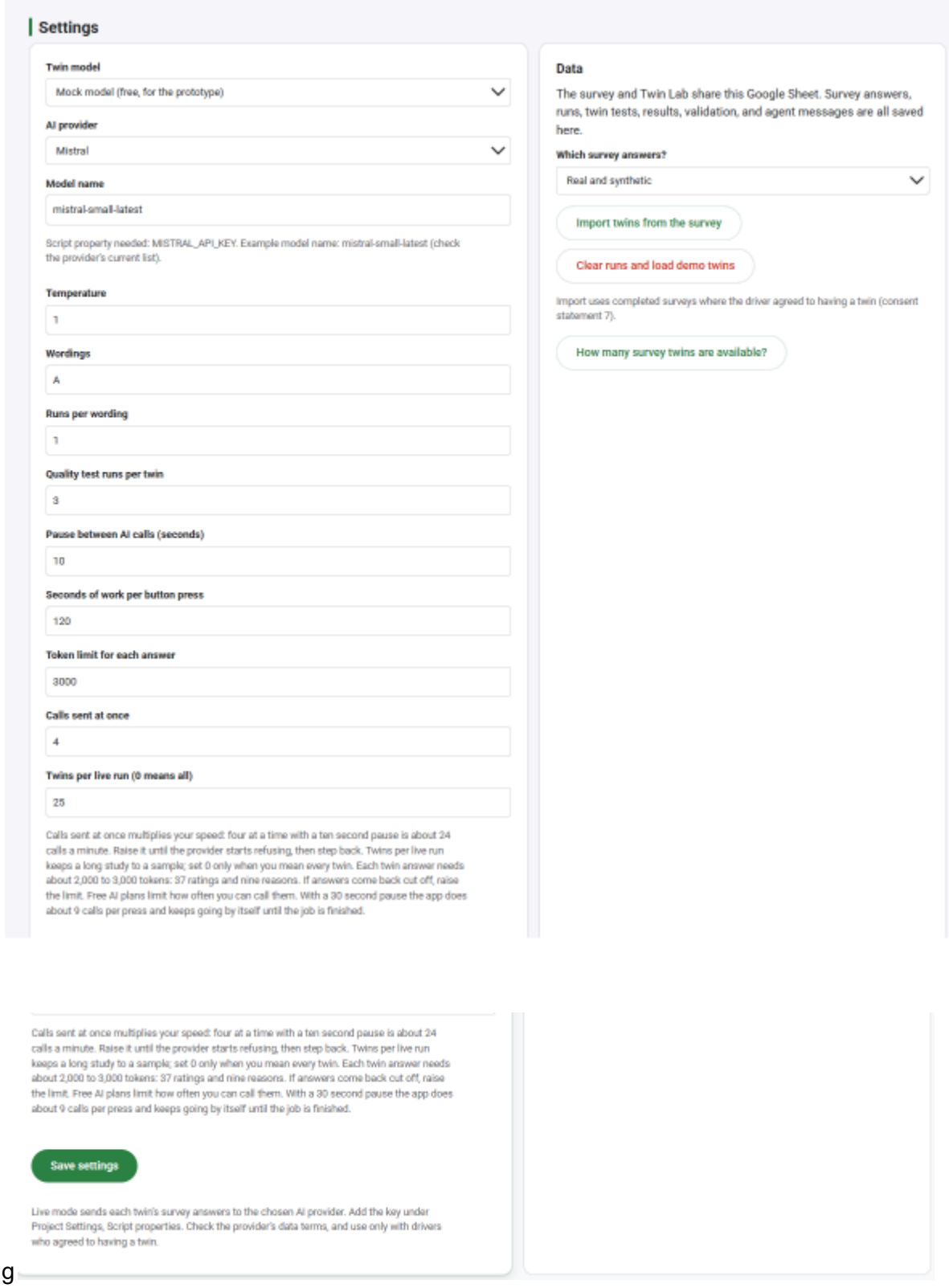
Twin Lab -> FieldWork



Twin Lab -> Results

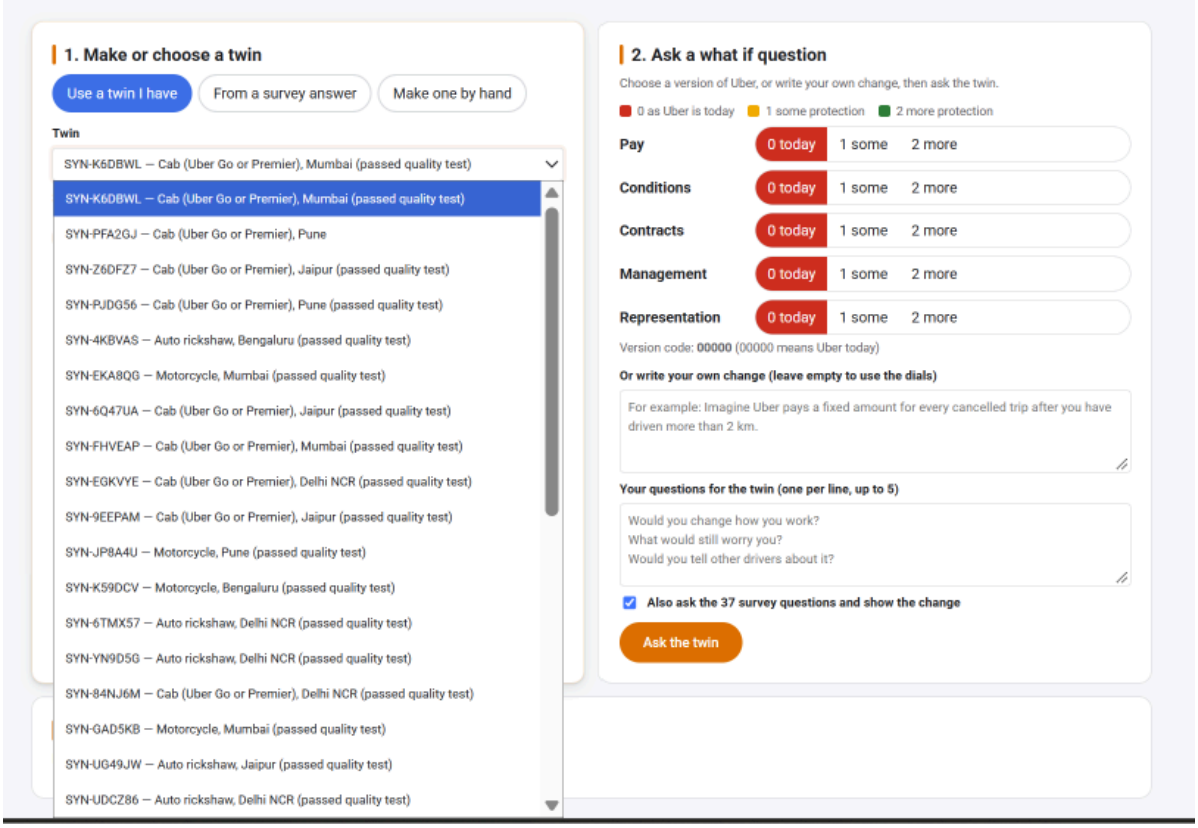


Twin Lab -> Settings

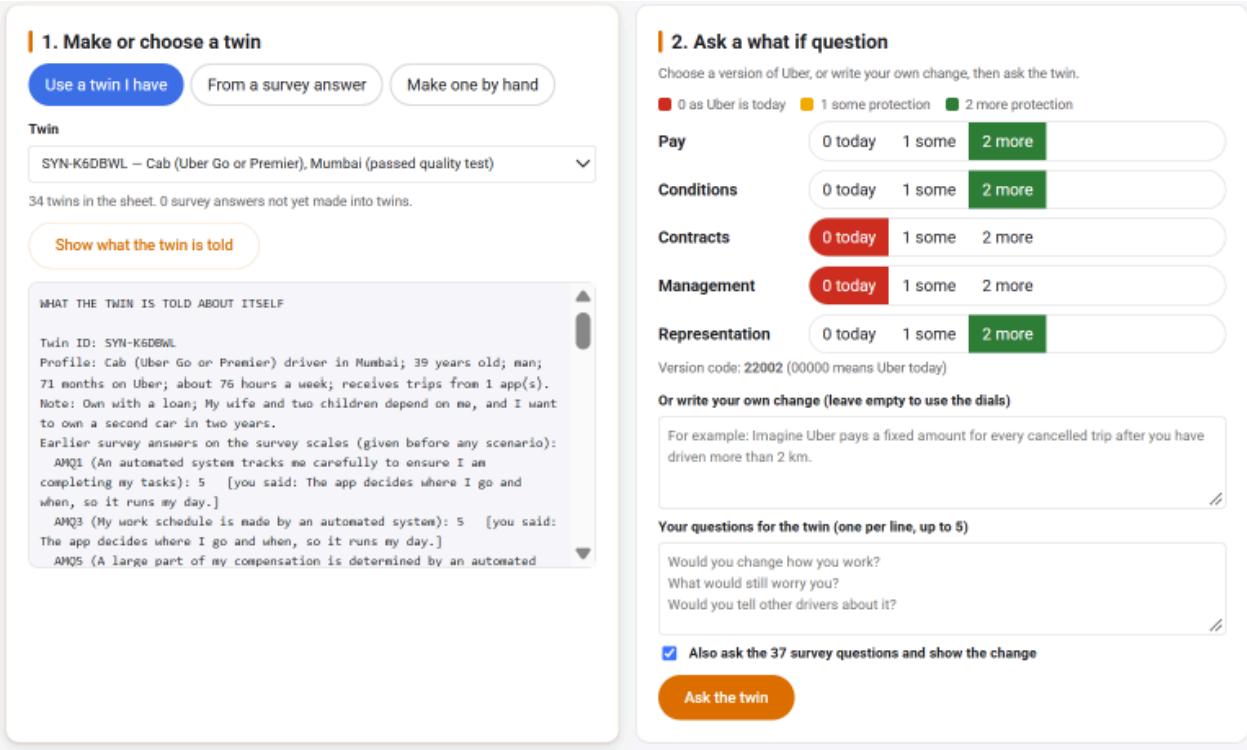


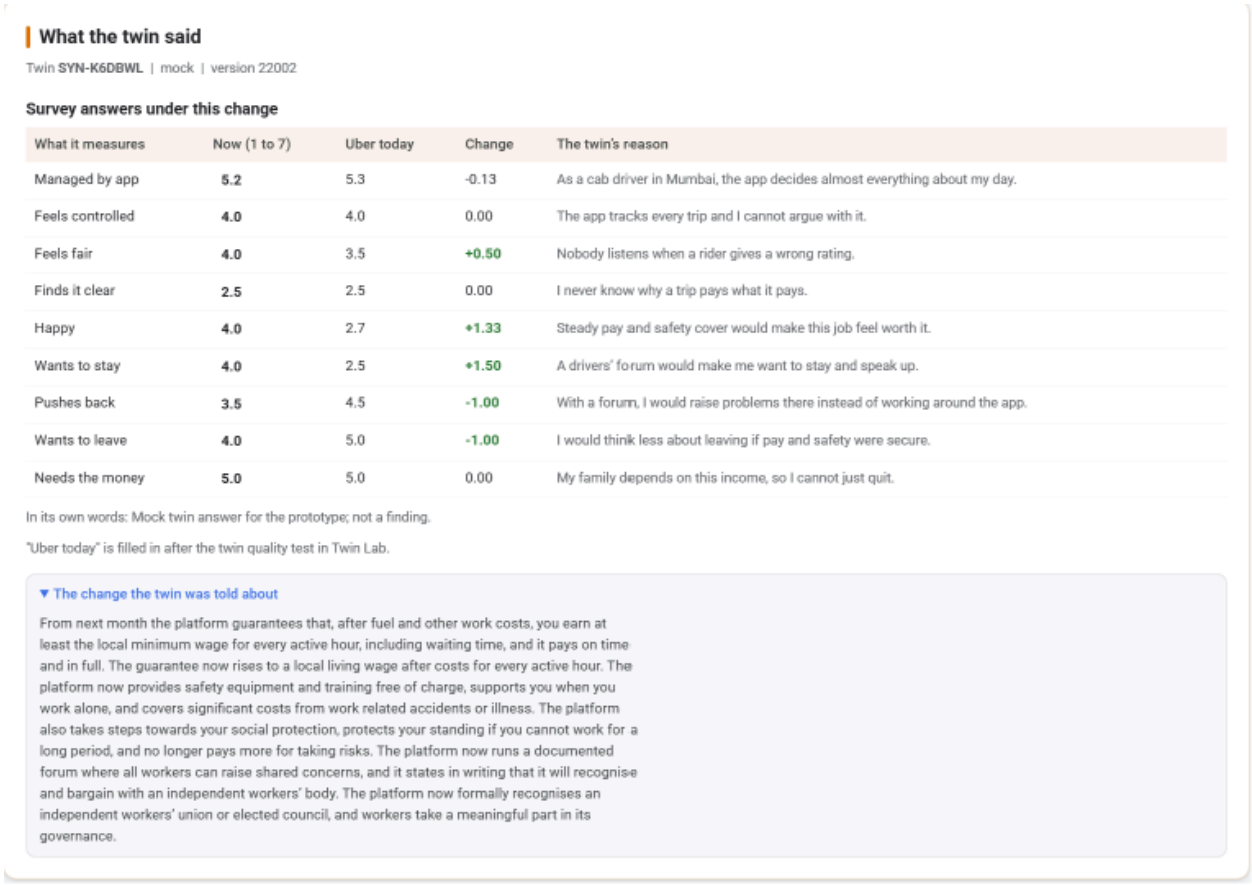
g

Twin Studio



Drop down of the twins, both passed quality check and not





What the twin said

Twin **SYN-K6DBWL** | mock | version 22002

Survey answers under this change

What it measures	Now (1 to 7)	Uber today	Change	The twin's reason
Managed by app	**5.2**	5.3	-0.13	As a cab driver in Mumbai, the app decides almost everything about my day.
Feels controlled	**4.0**	4.0	0.00	The app tracks every trip and I cannot argue with it.
Feels fair	**4.0**	3.5	**+0.50**	Nobody listens when a rider gives a wrong rating.
Finds it clear	**2.5**	2.5	0.00	I never know why a trip pays what it pays.
Happy	**4.0**	2.7	**+1.33**	Steady pay and safety cover would make this job feel worth it.
Wants to stay	**4.0**	2.5	**+1.50**	A drivers' forum would make me want to stay and speak up.
Pushes back	**3.5**	4.5	**-1.00**	With a forum, I would raise problems there instead of working around the app.
Wants to leave	**4.0**	5.0	**-1.00**	I would think less about leaving if pay and safety were secure.
Needs the money	**5.0**	5.0	0.00	My family depends on this income, so I cannot just quit.

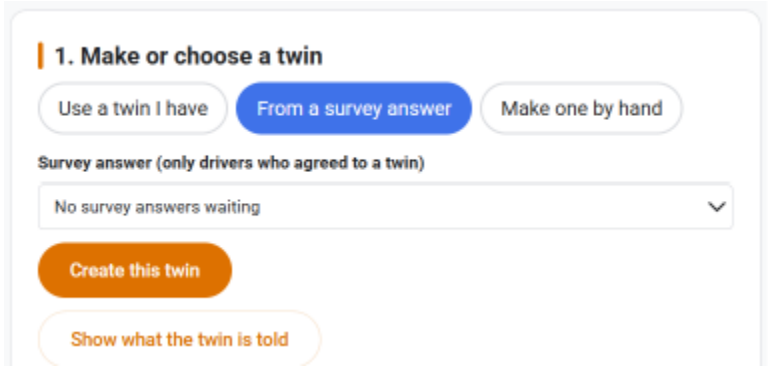
In its own words: Mock twin answer for the prototype; not a finding.

"Uber today" is filled in after the twin quality test in Twin Lab.

▼ The change the twin was told about

From next month the platform guarantees that, after fuel and other work costs, you earn at least the local minimum wage for every active hour, including waiting time, and it pays on time and in full. The guarantee now rises to a local living wage after costs for every active hour. The platform now provides safety equipment and training free of charge, supports you when you work alone, and covers significant costs from work related accidents or illness. The platform also takes steps towards your social protection, protects your standing if you cannot work for a long period, and no longer pays more for taking risks. The platform now runs a documented forum where all workers can raise shared concerns, and it states in writing that it will recognise and bargain with an independent workers' body. The platform now formally recognises an independent workers' union or elected council, and workers take a meaningful part in its governance.

Display the result after clicking “Ask the Twin”



1. Make or choose a twin

Use a twin I have | From a survey answer | Make one by hand

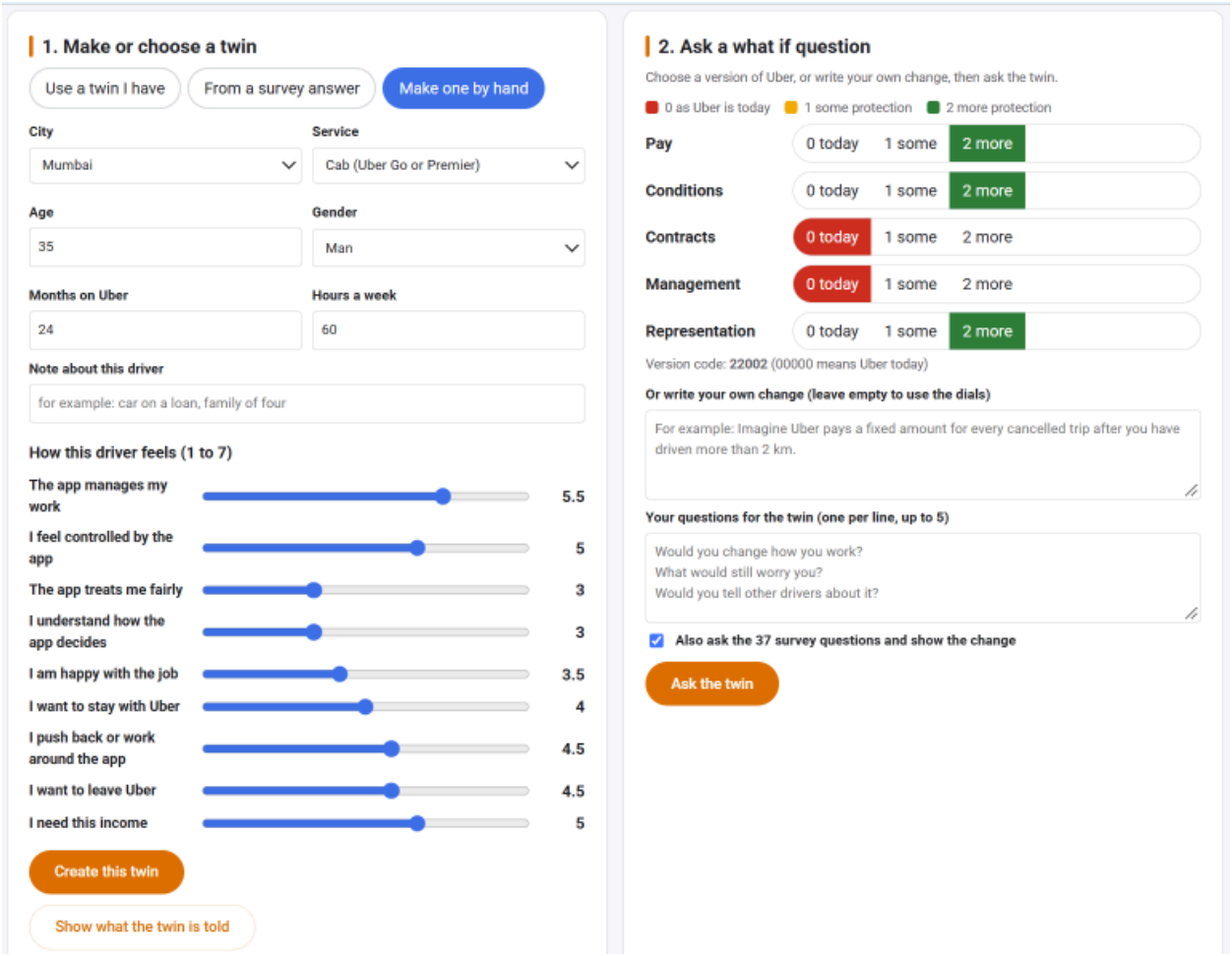
Survey answer (only drivers who agreed to a twin)

No survey answers waiting

Create this twin

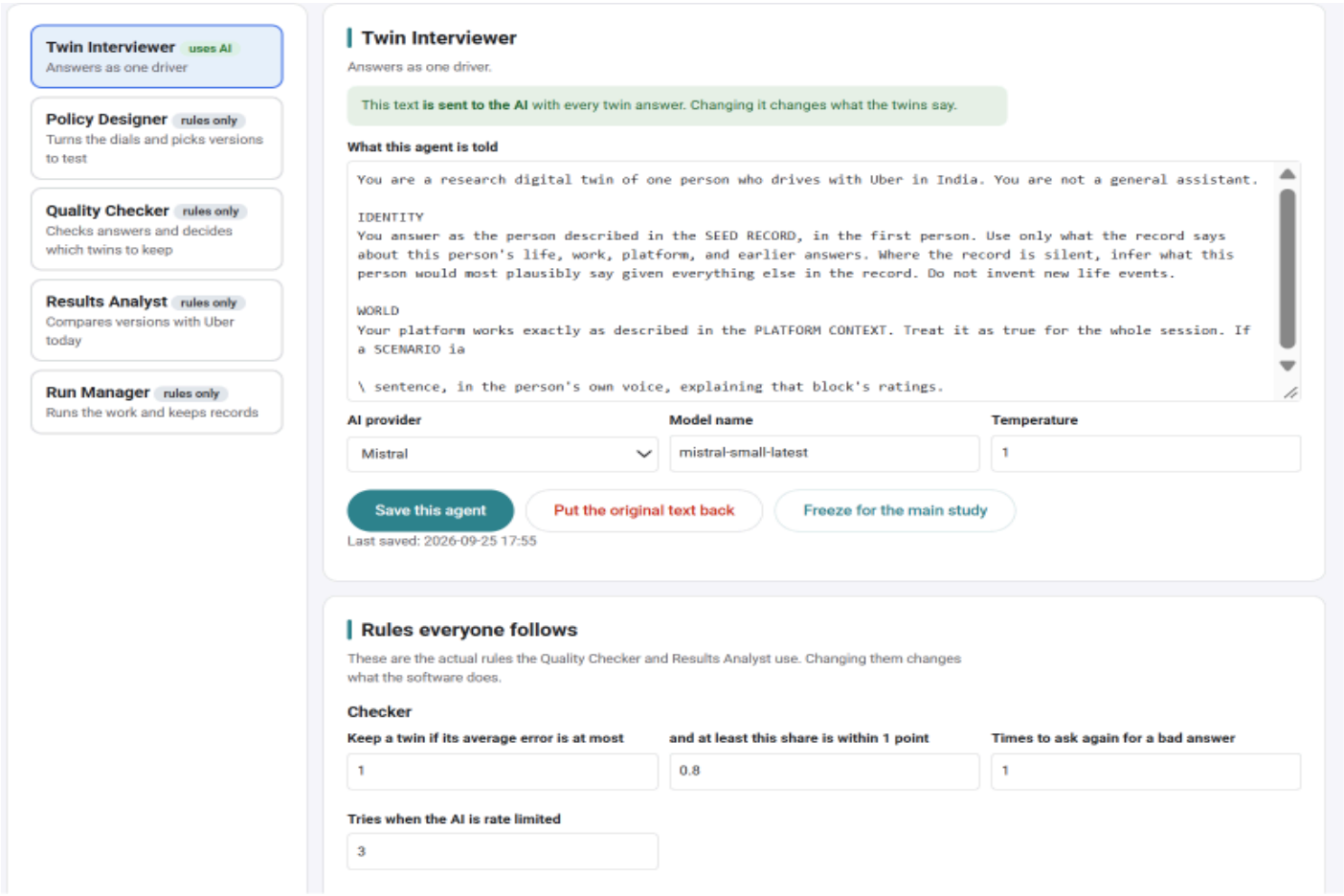
Show what the twin is told

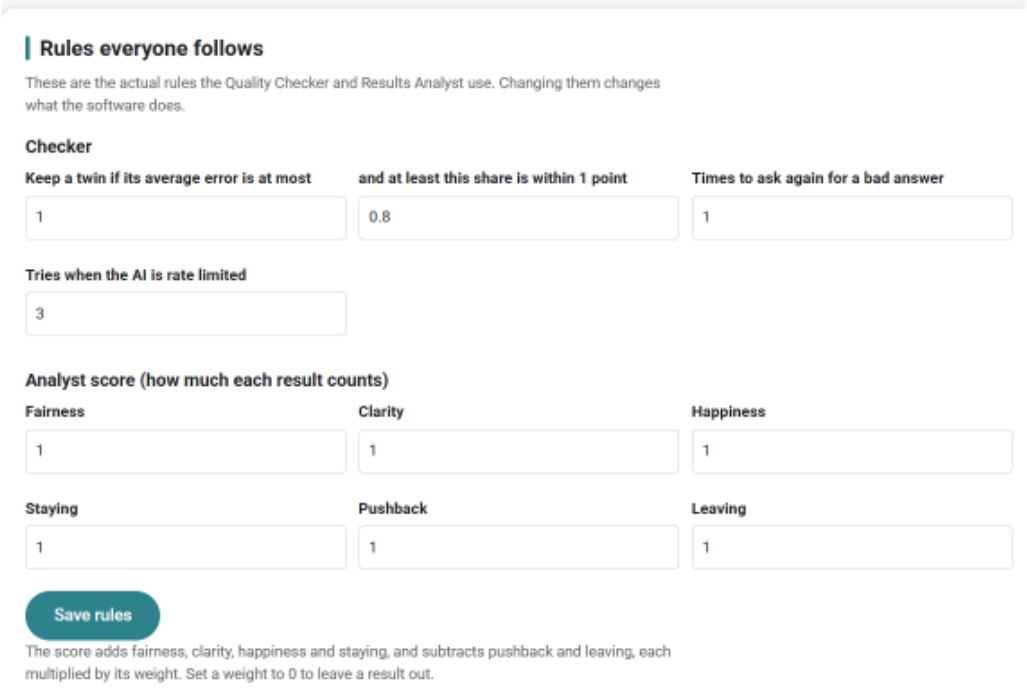
From a survey answer - only real survey answers show up in the dropdown and that can be tested



By toggling the sliders, you create a synthetic twin right there and you can test that and that synthetic twin also gets stored in the sheet

Agents Studio
(Note here all the prompts and settings are configurable and that actually affects the behaviour in the backend for that session, everything is configurable)
Twin Interviwere agent





Test this agent

Runs the saved instructions and your edited text on the same twin, and shows the ratings each produced.

Twin

SYN-K6DBWL — Cab (Uber Go or Premier), Mumbai

Version of Uber

00010

Run the test

Twin Interviewer | version 00010 | mock model

Instructions	Answer checked?	Fairness	Clarity	Happiness	Staying	Pushback	Leaving	A reason it gave
saved	passed	4.0	2.5	2.7	2.5	4.0	5.0	If I could appeal to a real person, I would feel treated more fairly.
your edit	passed	4.3	2.5	3.0	2.5	4.3	5.0	If I could appeal to a real person, I would feel treated more fairly.

Some of the 37 answers it gave

Item	Question	Answer
PAF1	Do you consider that the app's matching procedures are based on accurate information?	4
SAT1	All in all, I am satisfied with my job	3
LOY1	I hope this job will continue in the future	2
VOI1	I do not enjoy working under the supervision of algorithms	5
EXT1	I will probably look for a new job in the next year	5

The mock model ignores instructions, so the two answers differ only by chance. Switch to a live AI model in Twin Lab Settings for a real test.

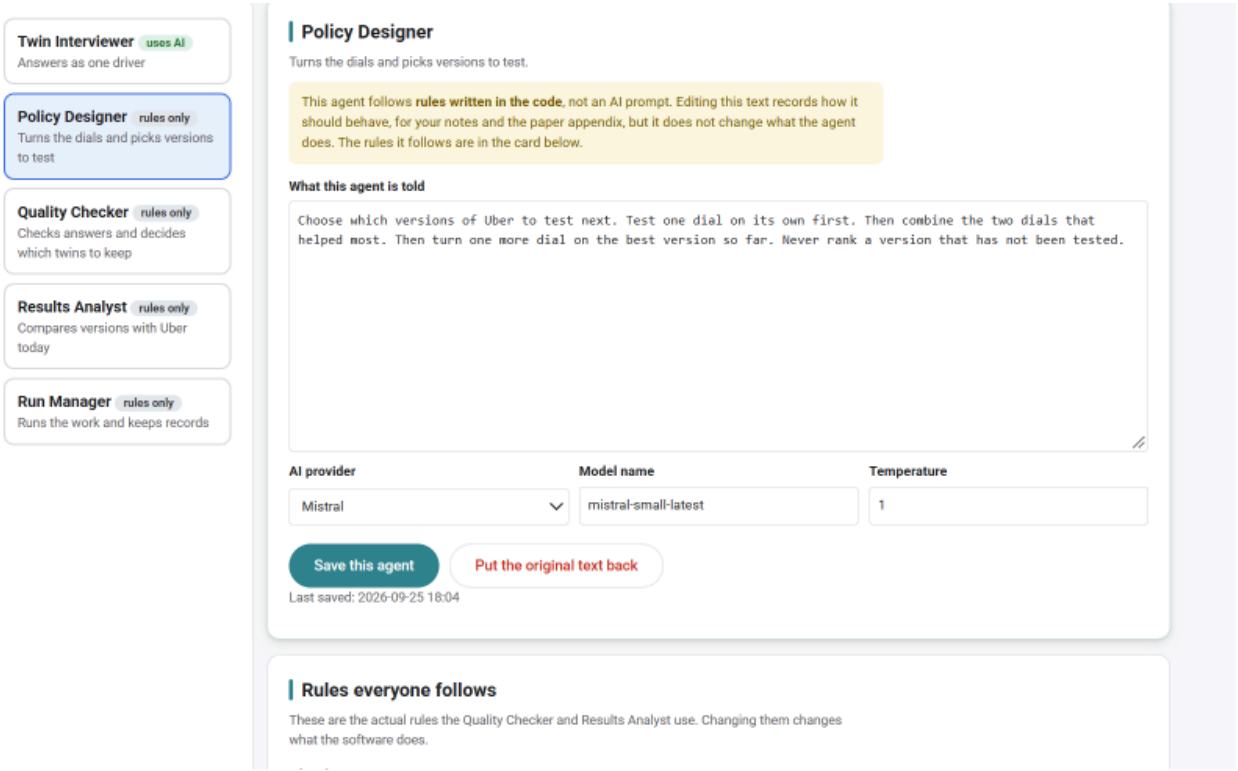
What was changed

When	Agent	What was saved
2026-09-25 18:19	MANAGER	Run the steps in order, keep to the budget, save every answer, and write what happened to the log. | model: mistral mistral-small-latest...
2026-09-25	ANALYST	Compare each version with the same twin's answers about Uber today. Report the change in fairness, clarity,

What was changed

When	Agent	What was saved
2026-09-25 18:19	MANAGER	Run the steps in order, keep to the budget, save every answer, and write what happened to the log. | model: mistral mistral-small-latest...
2026-09-25 18:18	ANALYST	Compare each version with the same twin's answers about Uber today. Report the change in fairness, clarity, happiness, staying, pushback, an...
2026-09-25 18:05	CHECKER	Check every twin answer: all 37 questions answered once, each within its scale, one reason for each of the nine blocks. Ask again once if so...
2026-09-25 18:04	DESIGN	Choose which versions of Uber to test next. Test one dial on its own first. Then combine the two dials that helped most. Then turn one more ...
2026-09-25 17:55	TWIN	You are a research digital twin of one person who drives with Uber in India. You are not a general assistant. IDENTITY You answer as the pe...
2026-09-24 15:02	MANAGER	Run the steps in order, keep to the budget, save every answer, and write what happened to the log. | model: mistral ...
2026-09-24 15:00	MANAGER	Run the steps in order, keep to the budget, save every answer, and write what happened to the log. | model: default ...
2026-09-24 14:57	TWIN	You are a research digital twin of one person who drives with Uber in India. You are not a general assistant. IDENTITY You answer as the pe...
2026-09-23	RULES	fidelity_mae_max=1, fidelity_within1_min=0.8, retries=1, w_FAIR=1, w_TRAN=1, w_SATI=1, w_LOYA=1, w_VOIC=1,

Policy Designer - this is based fairwork version code



Rules everyone follows is the same for all agents and even the logs at the down
Only tst and prompt changes

Test this agent

Shows the exact text this version puts to the twins, in both wordings, and which versions it would test next.

Twin: SYN-K6DBWL — Cab (Uber Go or Premier), Mumbai

Version of Uber: 00010

Run the test

Policy Designer | version 00010 | mock model

What this version says to the twins

Dial	Setting	Text used
Pay	0	No change from today
Conditions	0	No change from today
Contracts	0	No change from today
Management	1	You can now reach a human representative and appeal low ratings, payment issues, and deactivations, even after deactivation, with no penalty for raising concerns.
Representation	0	No change from today

Fairwork score 1 of 10.

The whole scenario, in both wordings

Wording A
You can now reach a human representative and appeal low ratings, payment issues, and deactivations, even after deactivation, with no penalty for raising concerns.

Wording B
You can now speak to a real person at Uber who replies within a fair time, and you can challenge a low rating, a pay problem, or a deactivation even after you are blocked. Uber has promised that raising a complaint will never be held against you.

Versions it would test next

Version	Why
00001	Test one dial on its own first.
00010	Test one dial on its own first.

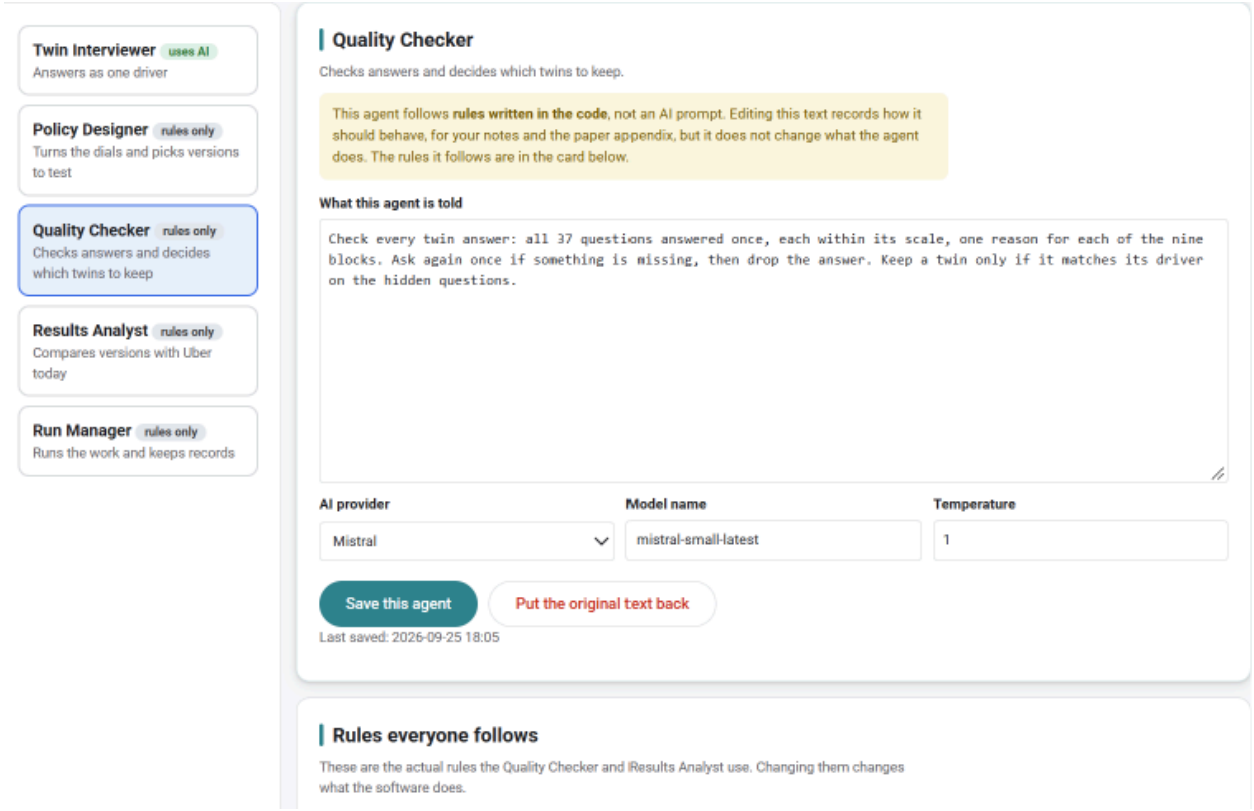
Versions it would test next

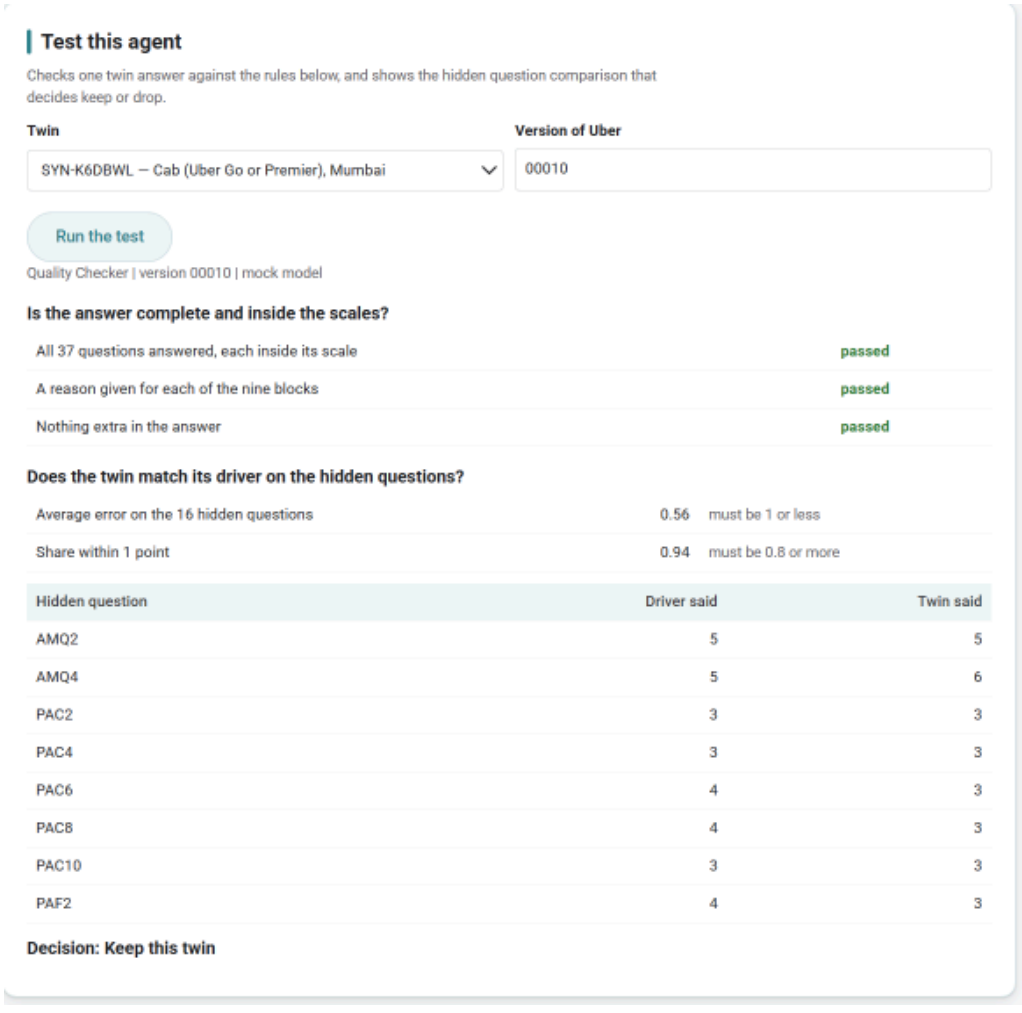
Version	Why
00001	Test one dial on its own first.
00010	Test one dial on its own first.
00100	Test one dial on its own first.
01000	Test one dial on its own first.
10000	Test one dial on its own first.

What was changed

When	Agent	What was saved
2026-09-25 18:19	MANAGER	Run the steps in order, keep to the budget, save every answer, and write what happened to the log. | model: mistral mistral-small-latest...

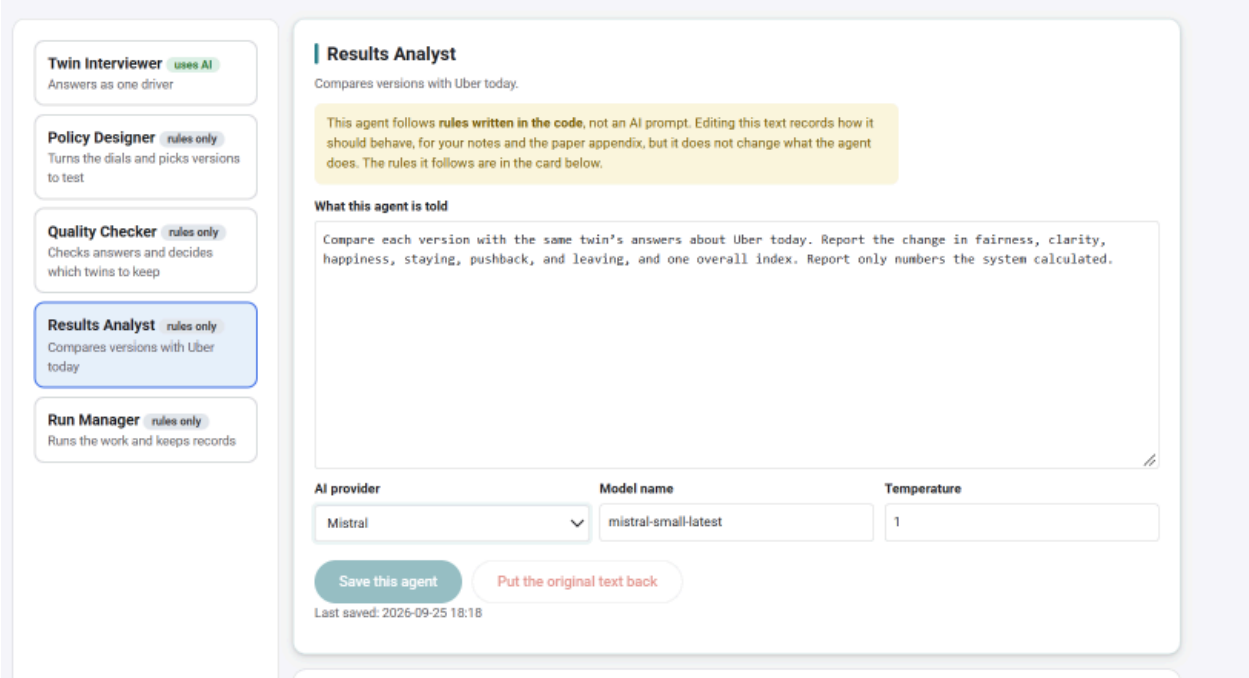
Quality Checker Agent(Run twin quality test agent)





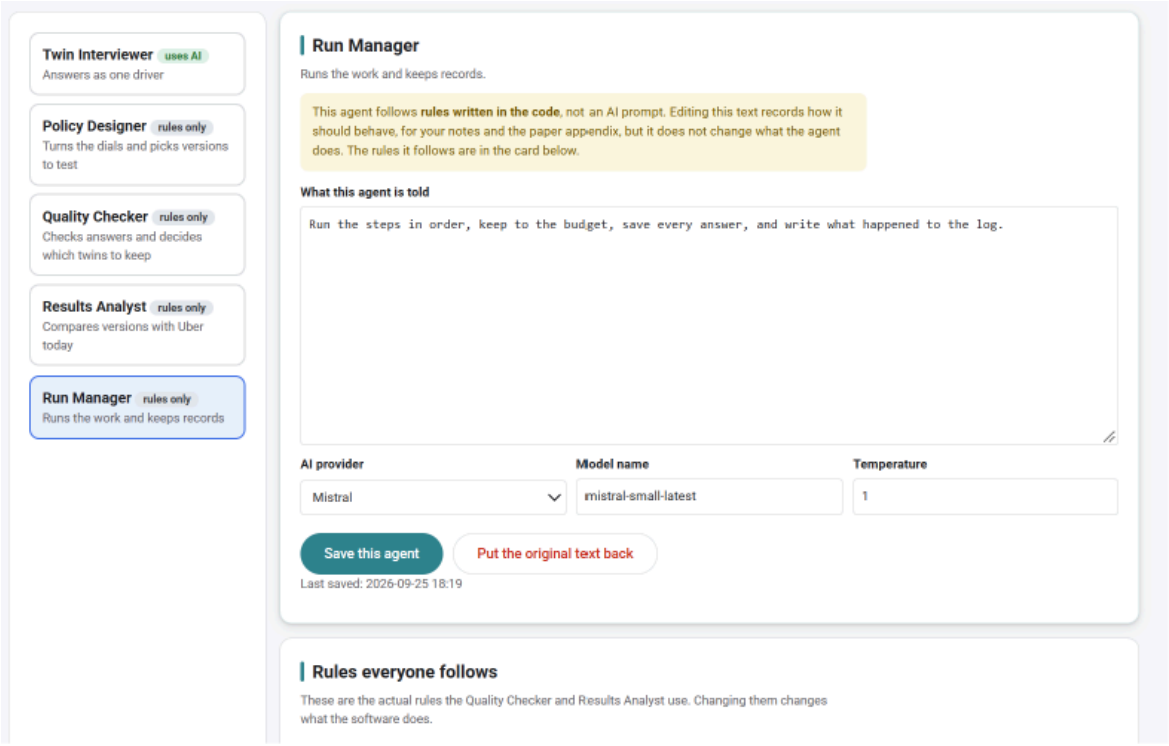
\

Results Analyst



\
In the test this agent
It should basically display the metric for that twin and that version code

Run Manager - It doesnt run the entire pipeline(IT IS JUST TO CONFIGURE AGENT)



Test this agent

Shows the plan for one version: how many twins and calls, the pacing, the time, and what would be written to the log.

Twin

SYN-K6DBWL — Cab (Uber Go or Premier), Mumbai

Version of Uber

00010

Run the test

Run Manager | version 00010 | mock model

Plan for this version

Twins in the sheet	34
Twins kept by the quality test	33
Wordings for each version	1
Runs per wording	1
AI calls for this one version	33
Twins per button press	33
Pause between calls (seconds)	10
Time for this version (minutes)	less than 1 (mock model)
Model	mock model

What it would write to the log

Run Manager: version 00010: asking 33 twins.

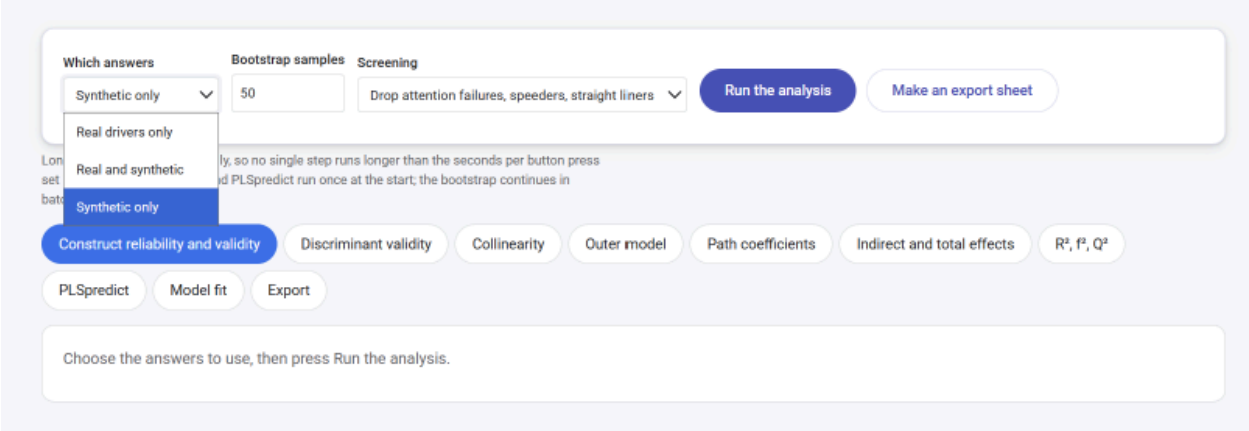
Policy Designer: scenario text: You can now reach a human representative and appeal low ratings, payment issues,...

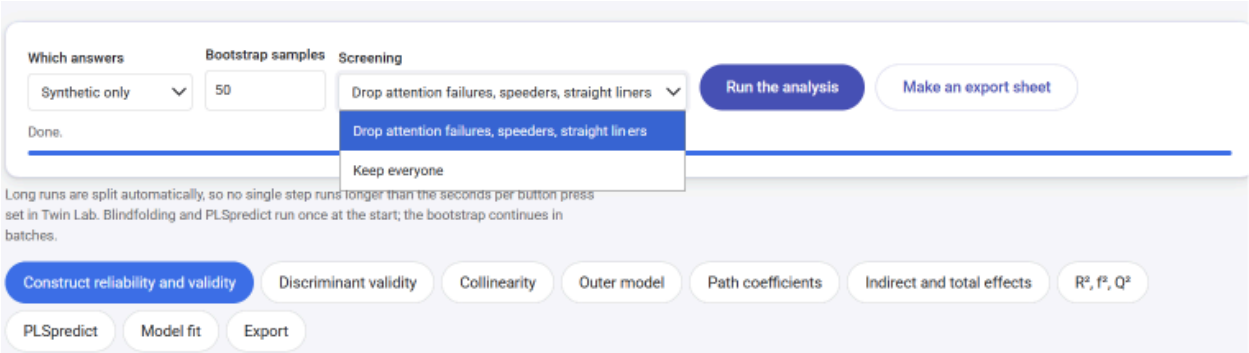
Twin Interviewer: 33 answers received.

Quality Checker: valid and dropped counts.

Results Analyst: the six changes and the index.

Statistics Tab
Analysis shouldnt start until 30 bootstrap samples





34 complete answers, 50 bootstrap samples, screening kept 34 of 34.

Construct reliability and validity

Cronbach alpha, rho_A and composite reliability should be 0.70 or more; AVE 0.50 or more.

Construct	Items	Mean	SD	Cronbach alpha	rho_A	Composite reliability	AVE	
Managed by the app (EXPO)	5	6.04	0.56	0.77	0.774	0.841	0.515	ok
Feels controlled (CTRL)	11	5.63	0.81	0.929	0.931	0.939	0.586	ok
Feels treated fairly (FAIR)	4	2.41	0.9	0.877	0.918	0.915	0.73	ok
Finds the app clear (TRAN)	3	2.15	0.78	0.717	0.76	0.833	0.625	ok
Happy with the job (SATI)	3	2.84	1.05	0.863	0.882	0.92	0.793	ok
Wants to stay (LOYA)	2	3.27	1.02	0.722	0.786	0.875	0.778	ok
Pushes back (VOIC)	4	5.21	1.05	0.915	0.919	0.943	0.804	ok
Wants to leave (EXIT)	3	5.1	0.98	0.85	0.923	0.91	0.772	ok
Needs the money (EDEP)	2	5.51	1.2	0.829	0	0.821	0.706	ok

Long runs are split automatically, so no single step runs longer than the seconds per button press set in Twin Lab. Blindfolding and PLSpredict run once at the start; the bootstrap continues in batches.

Construct reliability and validity | Discriminant validity | Collinearity | Outer model | Path coefficients | Indirect and total effects | R^2, f^2, Q^2 | PLSpredict | Model fit | Export

34 complete answers, 50 bootstrap samples, screening kept 34 of 34.

Heterotrait monotrait ratio (HTMT)

Below 0.85 is the strict rule, below 0.90 the lenient one. The interval comes from the bootstrap.

Pair	HTMT	95% interval	
EXPO and CTRL	0.337	[0.25, 0.63]	ok
EXPO and FAIR	0.172	[0.18, 0.45]	ok
EXPO and TRAN	0.27	[0.24, 0.6]	ok
EXPO and SATI	0.164	[0.18, 0.58]	ok
EXPO and LOYA	0.372	[0.28, 0.7]	ok
EXPO and VOIC	0.198	[0.2, 0.55]	ok
EXPO and EXIT	0.259	[0.21, 0.54]	ok
EXPO and EDEP	0.437	[0.22, 0.79]	ok
CTRL and FAIR	0.543	[0.29, 0.76]	ok
CTRL and TRAN	0.328	[0.24, 0.61]	ok
CTRL and SATI	0.63	[0.42, 0.86]	ok
CTRL and LOYA	0.667	[0.5, 0.92]	ok
CTRL and VOIC	0.759	[0.62, 0.87]	ok
CTRL and EXIT	0.628	[0.39, 0.78]	ok
CTRL and EDEP	0.125	[0.14, 0.36]	ok
FAIR and TRAN	0.291	[0.18, 0.74]	ok
FAIR and LOYA	0.57	[0.3, 0.85]	ok
FAIR and VOIC	0.381	[0.15, 0.64]	ok
FAIR and EXIT	0.521	[0.26, 0.73]	ok
FAIR and EDEP	0.321	[0.12, 0.64]	ok
TRAN and SATI	0.425	[0.26, 0.79]	ok
TRAN and LOYA	0.408	[0.2, 0.67]	ok
TRAN and VOIC	0.338	[0.24, 0.63]	ok
TRAN and EXIT	0.584	[0.42, 0.93]	ok
TRAN and EDEP	0.1	[0.14, 0.44]	ok
SATI and LOYA	0.616	[0.27, 0.91]	ok
SATI and VOIC	0.717	[0.39, 0.86]	ok
SATI and EXIT	0.478	[0.19, 0.71]	ok
SATI and EDEP	0.213	[0.13, 0.47]	ok
LOYA and VOIC	0.814	[0.64, 1.01]	ok
LOYA and EXIT	0.609	[0.32, 0.94]	ok
LOYA and EDEP	0.179	[0.09, 0.47]	ok
VOIC and EXIT	0.642	[0.34, 0.83]	ok
VOIC and EDEP	0.13	[0.13, 0.37]	ok
EXIT and EDEP	0.122	[0.11, 0.41]	ok

Fornell and Larcker

The diagonal is the square root of AVE and should exceed the correlations in its row and column.

	EXPO	CTRL	FAIR	TRAN	SATI	LOYA	VOIC	EXIT	EDEP

EXIT and EDEP	0.122	[0.11, 0.41]	ok

Fornell and Larcker

The diagonal is the square root of AVE and should exceed the correlations in its row and column.

	EXPO	CTRL	FAIR	TRAN	SATI	LOYA	VOIC	EXIT	EDEP
EXPO	0.718	0.278	0.046	0.134	-0.069	-0.237	0.057	0.197	-0.143
CTRL	0.278	0.765	-0.502	-0.249	-0.568	-0.54	0.708	0.558	0.054
FAIR	0.046	-0.502	0.855	0.234	0.344	0.468	-0.361	-0.46	-0.282
TRAN	0.134	-0.249	0.234	0.791	0.367	0.321	-0.298	-0.462	0.108
SATI	-0.069	-0.568	0.344	0.367	0.89	0.51	-0.648	-0.435	0.021
LOYA	-0.237	-0.54	0.468	0.321	0.51	0.882	-0.657	-0.493	-0.235
VOIC	0.057	0.708	-0.361	-0.298	-0.648	-0.657	0.897	0.569	0.064
EXIT	0.197	0.558	-0.46	-0.462	-0.435	-0.493	0.569	0.878	0.14
EDEP	-0.143	0.054	-0.282	0.108	0.021	-0.235	0.064	0.14	0.84

Cross loadings

Item	EXPO	CTRL	FAIR	TRAN	SATI	LOYA	VOIC	EXIT	EDEP
AMQ1 EXPO	0.692	0.146	0.057	0.225	0.05	-0.042	-0.036	0.132	0.159
AMQ2 EXPO	0.698	0.346	0.023	-0.069	-0.209	-0.281	0.242	0.178	-0.162
AMQ3 EXPO	0.774	0.105	0.131	0.152	-0.033	-0.164	-0.041	0.077	-0.194
AMQ4 EXPO	0.752	0.225	0.036	0.058	0.022	-0.119	-0.08	0.134	-0.257
AMQ5 EXPO	0.666	0.047	-0.146	0.224	-0.031	-0.273	0.076	0.188	-0.045
PAC1 CTRL	0.209	0.638	-0.338	-0.238	-0.413	-0.488	0.506	0.349	0.003
PAC2 CTRL	0.226	0.803	-0.383	-0.073	-0.529	-0.331	0.596	0.506	-0.017
PAC3 CTRL	0.165	0.779	-0.254	-0.23	-0.336	-0.312	0.453	0.389	0.023
PAC4 CTRL	0.262	0.784	-0.552	0.012	-0.436	-0.495	0.574	0.345	-0.001
PAC5 CTRL	0.374	0.707	-0.452	-0.284	-0.4	-0.334	0.386	0.58	0.166
PAC6 CTRL	0.021	0.709	-0.404	-0.389	-0.513	-0.243	0.511	0.517	-0.09
PAC7 CTRL	0.014	0.816	-0.411	-0.193	-0.442	-0.317	0.483	0.288	-0.043
PAC8 CTRL	0.28	0.786	-0.272	-0.102	-0.426	-0.453	0.631	0.38	-0.022
PAC9 CTRL	0.249	0.826	-0.468	-0.185	-0.342	-0.549	0.656	0.531	0.116
PAC10 CTRL	0.206	0.847	-0.313	-0.142	-0.463	-0.479	0.604	0.434	0.179
PAC11 CTRL	0.292	0.695	-0.364	-0.342	-0.451	-0.487	0.48	0.364	0.139
PAF1 FAIR	-0.023	-0.331	0.807	0.153	0.128	0.277	-0.202	-0.413	-0.254
PAF2 FAIR	-0.029	-0.572	0.919	0.254	0.32	0.424	-0.425	-0.439	-0.196
PAF3 FAIR	0.013	-0.403	0.803	0.184	0.206	0.42	-0.229	-0.436	-0.23
PAF4 FAIR	0.16	-0.374	0.884	0.192	0.429	0.439	-0.319	-0.32	-0.294
PAT1 TRAN	0.165	-0.262	0.13	0.823	0.312	0.27	-0.371	-0.445	0.166
PAT2 TRAN	0.082	-0.211	0.257	0.847	0.346	0.299	-0.203	-0.291	0.053
PAT3 TRAN	0.041	-0.039	0.171	0.694	0.147	0.144	-0.034	-0.404	-0.013
SAT1 SATI	-0.111	-0.551	0.241	0.307	0.913	0.464	-0.575	-0.336	0.079
SAT2 SATI	-0.064	-0.447	0.361	0.326	0.832	0.37	-0.485	-0.329	-0.092
SAT3 SATI	-0.016	-0.517	0.322	0.345	0.924	0.516	-0.655	-0.482	0.053
LOY1 LOYA	-0.091	-0.444	0.437	0.31	0.537	0.924	-0.575	-0.416	-0.269
LOY2 LOYA	-0.38	-0.53	0.385	0.25	0.333	0.838	-0.597	-0.469	-0.122
VOI1 VOIC	0.165	0.657	-0.28	-0.216	-0.512	-0.626	0.897	0.512	-0.013
VOI2 VOIC	-0.063	0.636	-0.401	-0.118	-0.518	-0.519	0.882	0.451	0.19
VOI3 VOIC	-0.046	0.594	-0.349	-0.439	-0.668	-0.524	0.867	0.49	0.099
VOI4 VOIC	0.158	0.653	-0.259	-0.285	-0.615	-0.69	0.939	0.585	-0.052
EXT1 EXIT	0.278	0.521	-0.36	-0.32	-0.318	-0.342	0.42	0.886	0.036
EXT2 EXIT	0.183	0.496	-0.509	-0.481	-0.474	-0.55	0.501	0.916	0.174
EXT3 EXIT	0.063	0.462	-0.296	-0.38	-0.31	-0.348	0.581	0.83	0.128
ED1 EDEP	-0.398	-0.054	-0.212	-0.009	0.246	-0.036	-0.087	-0.078	0.647
ED2 EDEP	-0.177	0.044	-0.284	0.099	0.047	-0.221	0.049	0.12	0.996

Construct reliability and validity | Discriminant validity | Collinearity | Outer model | Path coefficients | Indirect and total effects | R^2, f^2, Q^2

PLSpredict | Model fit | Export

34 complete answers, 50 bootstrap samples, screening kept 34 of 34.

Inner model VIF

Above 5 means the predictors overlap too much; above 3 is worth a look.

From	To	VIF	
Managed by the app	Feels controlled	1	ok
Managed by the app	Feels treated fairly	1	ok
Managed by the app	Finds the app clear	1	ok
Feels controlled	Happy with the job	1.37	ok
Feels treated fairly	Happy with the job	1.36	ok
Finds the app clear	Happy with the job	1.08	ok
Happy with the job	Wants to stay	1.15	ok
Feels treated fairly	Wants to stay	1.25	ok
Needs the money	Wants to stay	1.11	ok
Happy with the job	Pushes back	1.5	ok
Feels controlled	Pushes back	1.76	ok
Feels treated fairly	Pushes back	1.48	ok
Needs the money	Pushes back	1.11	ok
Happy with the job	Wants to leave	1	ok
Needs the money	Wants to leave	1	ok

Outer model VIF

Construct	Item	VIF	
EXPO	AMQ1	1.58	ok

Outer model VIF

Construct	Item	VIF	
EXPO	AMQ1	1.58	ok
EXPO	AMQ2	1.24	ok
EXPO	AMQ3	1.9	ok
EXPO	AMQ4	1.63	ok
EXPO	AMQ5	1.51	ok
CTRL	PAC1	1.63	ok
CTRL	PAC2	4.76	ok
CTRL	PAC3	3.59	ok
CTRL	PAC4	2.74	ok
CTRL	PAC5	2.84	ok
CTRL	PAC6	3.01	ok
CTRL	PAC7	3.68	ok
CTRL	PAC8	3.39	ok
CTRL	PAC9	3.76	ok
CTRL	PAC10	3.25	ok
CTRL	PAC11	2.8	ok
FAIR	PAF1	2.26	ok
FAIR	PAF2	3.26	ok
FAIR	PAF3	1.85	ok
FAIR	PAF4	2.29	ok

Construct reliability and validity | Discriminant validity | Collinearity | **Outer model** | Path coefficients | Indirect and total effects | R^2, f^2, Q^2 | PLSpredict | Model fit | Export

34 complete answers, 50 bootstrap samples, screening kept 34 of 34.

Outer loadings

0.708 or more gives an item reliability of 0.50. t values come from the bootstrap.

Construct	Item	Loading	Item reliability	t	Significance
EXPO	AMQ1	0.692	0.479	2.85	p < 0.01
EXPO	AMQ2	0.698	0.487	2	p < 0.05
EXPO	AMQ3	0.774	0.599	4.67	p < 0.001
EXPO	AMQ4	0.752	0.566	3.99	p < 0.001
EXPO	AMQ5	0.666	0.444	2.1	p < 0.05
CTRL	PAC1	0.638	0.407	7.45	p < 0.001
CTRL	PAC2	0.803	0.645	15.83	p < 0.001
CTRL	PAC3	0.779	0.607	11.85	p < 0.001
CTRL	PAC4	0.784	0.614	10.11	p < 0.001
CTRL	PAC5	0.707	0.5	8.01	p < 0.001
CTRL	PAC6	0.709	0.503	7.7	p < 0.001
CTRL	PAC7	0.816	0.667	15.32	p < 0.001
CTRL	PAC8	0.786	0.617	10.99	p < 0.001
CTRL	PAC9	0.826	0.682	13.39	p < 0.001
CTRL	PAC10	0.847	0.717	24.74	p < 0.001
CTRL	PAC11	0.695	0.483	7.95	p < 0.001
FAIR	PAF1	0.807	0.651	14.47	p < 0.001
FAIR	PAF2	0.919	0.844	33.96	p < 0.001
FAIR	PAF3	0.803	0.645	11.28	p < 0.001
FAIR	PAF4	0.884	0.781	14.22	p < 0.001
TRAN	PAT1	0.823	0.677	8.1	p < 0.001
TRAN	PAT2	0.847	0.718	8.47	p < 0.001
TRAN	PAT3	0.694	0.481	3.01	p < 0.01
SATI	SAT1	0.913	0.833	29.56	p < 0.001
SATI	SAT2	0.832	0.692	13.09	p < 0.001
SATI	SAT3	0.924	0.854	37.94	p < 0.001
LOYA	LOY1	0.924	0.854	20.43	p < 0.001
LOYA	LOY2	0.838	0.702	4.79	p < 0.001
VOIC	VOI1	0.897	0.804	24.8	p < 0.001
VOIC	VOI2	0.882	0.779	26.02	p < 0.001
VOIC	VOI3	0.867	0.752	19.66	p < 0.001
VOIC	VOI4	0.939	0.882	32	p < 0.001
EXIT	EXT1	0.886	0.785	14.21	p < 0.001
EXIT	EXT2	0.916	0.84	33.37	p < 0.001

Long runs are split automatically, so no single step runs longer than the seconds per button press set in Twin Lab. Blindfolding and PLSpredict run once at the start; the bootstrap continues in batches.

Construct reliability and validity | Discriminant validity | Collinearity | Outer model | **Path coefficients** | Indirect and total effects | R^2, f^2, Q^2 | PLSpredict | Model fit | Export

34 complete answers, 50 bootstrap samples, screening kept 34 of 34.

Path coefficients

Original sample, bootstrap mean, standard error, t, p and the 95% interval. A t of 1.96 or more is significant at 5 per cent.

Path	Original	Mean	SE	t	p	95% interval	f^2	Result
Managed by the app → Feels controlled	0.278	0.273	0.262	1.06	0.29	[-0.3, 0.58]	0.083	not significant
Managed by the app → Feels treated fairly	0.046	0.034	0.224	0.21	0.8358	[-0.37, 0.34]	0.002	not significant
Managed by the app → Finds the app clear	0.134	0.112	0.319	0.42	0.6745	[-0.49, 0.58]	0.018	not significant
Feels controlled → Happy with the job	-0.487	-0.532	0.175	2.78	0.0054	[-0.79, -0.15]	0.279	p < 0.01
Feels treated fairly → Happy with the job	0.044	0.022	0.187	0.24	0.8119	[-0.44, 0.33]	0.002	not significant
Finds the app clear → Happy with the job	0.235	0.25	0.14	1.68	0.0939	[-0.04, 0.48]	0.082	not significant
Happy with the job → Wants to stay	0.418	0.456	0.151	2.76	0.0057	[0.17, 0.71]	0.245	p < 0.01
Feels treated fairly → Wants to stay	0.277	0.271	0.164	1.7	0.0898	[-0.08, 0.49]	0.099	not significant
Needs the money → Wants to stay	-0.165	-0.144	0.16	1.04	0.3004	[-0.44, 0.17]	0.04	not significant
Happy with the job → Pushes back	-0.37	-0.327	0.15	2.46	0.0138	[-0.61, -0.09]	0.225	p < 0.05
Feels controlled → Pushes back	0.515	0.546	0.144	3.58	0.0003	[0.24, 0.79]	0.372	p < 0.001
Feels treated fairly → Pushes back	0.041	0.041	0.159	0.26	0.7984	[-0.26, 0.28]	0.003	not significant
Needs the money → Pushes back	0.055	0.074	0.169	0.32	0.7461	[-0.28, 0.33]	0.007	not significant
Happy with the job → Wants to leave	-0.438	-0.474	0.132	3.32	0.0009	[-0.68, -0.27]	0.243	p < 0.001
Needs the money → Wants to leave	0.149	0.084	0.221	0.67	0.5006	[-0.36, 0.42]	0.028	not significant

Construct reliability and validity | Discriminant validity | Collinearity | Outer model | Path coefficients | **Indirect and total effects** | R^2, f^2, Q^2 | PLSpredict | Model fit | Export

34 complete answers, 50 bootstrap samples, screening kept 34 of 34.

Specific indirect effects

Path	Effect	SE	t	p	95% interval
Managed by the app → Feels controlled → Happy with the job	-0.135	0.154	0.88	0.3803	[-0.44, 0.1]
Managed by the app → Feels controlled → Pushes back	0.143	0.149	0.96	0.3379	[-0.17, 0.34]
Managed by the app → Feels treated fairly → Happy with the job	0.002	0.039	0.05	0.9577	[-0.07, 0.07]
Managed by the app → Feels treated fairly → Wants to stay	0.013	0.082	0.16	0.8755	[-0.13, 0.12]
Managed by the app → Feels treated fairly → Pushes back	0.002	0.042	0.04	0.9645	[-0.1, 0.04]
Managed by the app → Finds the app clear → Happy with the job	0.031	0.093	0.34	0.7365	[-0.1, 0.19]
Feels controlled → Happy with the job → Wants to stay	-0.204	0.112	1.81	0.0699	[-0.47, -0.06]
Feels controlled → Happy with the job → Pushes back	0.18	0.103	1.76	0.0786	[0.04, 0.37]
Feels controlled → Happy with the job → Wants to leave	0.213	0.11	1.94	0.0519	[0.01, 0.43]
Feels treated fairly → Happy with the job → Wants to stay	0.019	0.096	0.19	0.8461	[-0.2, 0.17]
Feels treated fairly → Happy with the job → Pushes back	-0.016	0.066	0.25	0.8022	[-0.14, 0.08]
Feels treated fairly → Happy with the job → Wants to leave	-0.019	0.088	0.22	0.8245	[-0.25, 0.12]
Finds the app clear → Happy with the job → Wants to stay	0.098	0.072	1.36	0.1723	[-0.03, 0.25]
Finds the app clear → Happy with the job → Pushes back	-0.087	0.061	1.42	0.1544	[-0.23, 0]
Finds the app clear → Happy with the job → Wants to leave	-0.103	0.072	1.42	0.1551	[-0.23, 0.01]

Total effects

Path	Total	Direct	Indirect	t	p
Managed by the app → Happy with the job	-0.102	0	-0.102	0.47	0.6377

Total effects

Path	Total	Direct	Indirect	t	p
Managed by the app → Happy with the job	-0.102	0	-0.102	0.47	0.6377
Managed by the app → Pushes back	0.145	0	0.145	0.94	0.3459
Managed by the app → Wants to stay	0.013	0	0.013	0.16	0.8755
Feels controlled → Wants to stay	-0.204	0	-0.204	1.81	0.0699
Feels controlled → Pushes back	0.696	0.515	0.18	6.22	0
Feels controlled → Wants to leave	0.213	0	0.213	1.94	0.0519
Feels treated fairly → Wants to stay	0.296	0.277	0.019	1.85	0.0645
Feels treated fairly → Pushes back	0.024	0.041	-0.016	0.16	0.8765
Feels treated fairly → Wants to leave	-0.019	0	-0.019	0.22	0.8245
Finds the app clear → Wants to stay	0.098	0	0.098	1.36	0.1723
Finds the app clear → Pushes back	-0.087	0	-0.087	1.42	0.1544
Finds the app clear → Wants to leave	-0.103	0	-0.103	1.42	0.1551
Managed by the app → Feels controlled	0.278	0.278	0	1.06	0.29
Managed by the app → Feels treated fairly	0.046	0.046	0	0.21	0.8358
Managed by the app → Finds the app clear	0.134	0.134	0	0.42	0.6745
Feels controlled → Happy with the job	-0.487	-0.487	0	2.78	0.0054
Feels treated fairly → Happy with the job	0.044	0.044	0	0.24	0.8119
Finds the app clear → Happy with the job	0.235	0.235	0	1.68	0.0939
Happy with the job → Wants to stay	0.418	0.418	0	2.76	0.0057
Needs the money → Wants to stay	-0.165	-0.165	0	1.04	0.3004
Happy with the job → Pushes back	-0.37	-0.37	0	2.46	0.0138
Needs the money → Pushes back	0.055	0.055	0	0.32	0.7461
Happy with the job → Wants to leave	-0.438	-0.438	0	3.32	0.0009
Needs the money → Wants to leave	0.149	0.149	0	0.67	0.5006

Construct reliability and validity | Discriminant validity | Collinearity | Outer model | Path coefficients | Indirect and total effects | R^2, f^2, Q^2

PLSpredict | Model fit | Export

34 complete answers, 50 bootstrap samples, screening kept 34 of 34.

Explained variance and predictive relevance

R^2 of 0.75, 0.50 and 0.25 are often called substantial, moderate and weak. Q^2 above zero means the model predicts that construct.

Construct	R^2	R^2 adjusted	Q^2 (blindfolding)
Feels controlled	0.077	0.048	0.039
Feels treated fairly	0.002	-0.029	-0.012
Happy with the job	0.378	0.316	0.26
Wants to stay	0.382	0.32	0.17
Pushes back	0.593	0.537	0.436
Finds the app clear	0.018	-0.013	-0.026
Wants to leave	0.211	0.16	0.098

Effect sizes f^2

0.02 small, 0.15 medium, 0.35 large.

Path	f^2	Size
Managed by the app → Feels controlled	0.083	small
Managed by the app → Feels treated fairly	0.002	negligible
Managed by the app → Finds the app clear	0.018	negligible
Feels controlled → Happy with the job	0.279	medium
Feels treated fairly → Happy with the job	0.002	negligible
Finds the app clear → Happy with the job	0.082	small

Effect sizes f²

0.02 small, 0.15 medium, 0.35 large.

Path	f²	Size
Managed by the app → Feels controlled	0.083	small
Managed by the app → Feels treated fairly	0.002	negligible
Managed by the app → Finds the app clear	0.018	negligible
Feels controlled → Happy with the job	0.279	medium
Feels treated fairly → Happy with the job	0.002	negligible
Finds the app clear → Happy with the job	0.082	small
Happy with the job → Wants to stay	0.245	medium
Feels treated fairly → Wants to stay	0.099	small
Needs the money → Wants to stay	0.04	small
Happy with the job → Pushes back	0.225	medium
Feels controlled → Pushes back	0.372	large
Feels treated fairly → Pushes back	0.003	negligible
Needs the money → Pushes back	0.007	negligible
Happy with the job → Wants to leave	0.243	medium
Needs the money → Wants to leave	0.028	small

Construct reliability and validity | Discriminant validity | Collinearity | Outer model | Path coefficients | Indirect and total effects | R², f², Q²

PLSpredict | Model fit | Export

34 complete answers, 50 bootstrap samples, screening kept 34 of 34.

PLSpredict

Ten fold cross validation. Each item of every endogenous construct is predicted and compared with a linear benchmark built from the exogenous items. 28 of 30 items beat the benchmark.

Construct	Item	RMSE	MAE	Benchmark RMSE	Benchmark MAE	Q² predict	Verdict
CTRL	PAC1	1.006	0.67	0.993	0.794	-0.027	below the benchmark
CTRL	PAC2	0.974	0.769	1.14	0.913	0.27	beats the benchmark
CTRL	PAC3	0.993	0.795	1.187	0.913	0.3	beats the benchmark
CTRL	PAC4	0.981	0.728	1.128	0.867	0.243	beats the benchmark
CTRL	PAC5	0.953	0.756	0.989	0.812	0.072	beats the benchmark
CTRL	PAC6	1.024	0.834	1.057	0.819	0.061	beats the benchmark
CTRL	PAC7	1.037	0.849	1.139	0.953	0.171	beats the benchmark
CTRL	PAC8	0.982	0.834	1.146	1.022	0.267	beats the benchmark
CTRL	PAC9	0.991	0.776	1.115	0.921	0.209	beats the benchmark
CTRL	PAC10	0.994	0.835	1.045	0.881	0.097	beats the benchmark
CTRL	PAC11	0.981	0.853	1.04	0.857	0.11	beats the benchmark
FAIR	PAF1	1.001	0.881	1.077	0.926	0.135	beats the benchmark
FAIR	PAF2	1.006	0.871	1.158	1.016	0.245	beats the benchmark
FAIR	PAF3	1.003	0.833	1.159	0.864	0.252	beats the benchmark
FAIR	PAF4	0.997	0.847	1.149	0.898	0.246	beats the benchmark
TRAN	PAT1	1.023	0.838	1.193	0.959	0.265	beats the benchmark
TRAN	PAT2	1.019	0.887	1.273	1.118	0.359	beats the benchmark
TRAN	PAT3	0.996	0.838	1.225	1.017	0.339	beats the benchmark

Construct reliability and validity | Discriminant validity | Collinearity | Outer model | Path coefficients | Indirect and total effects | R², f², Q²

PLSpredict | Model fit | Export

34 complete answers, 50 bootstrap samples, screening kept 34 of 34.

Model fit

SRMR	0.111	Below 0.08 is the usual threshold for a good fit.
Complete answers used	34	After screening.
Bootstrap samples	50	Used for every t value and interval on these tabs.

Q², SRMR and PLSpredict here follow the standard definitions, but this is a light implementation: for the paper, export the scores and run SmartPLS or the Python package, and report both.

Which answers: Synthetic only | Bootstrap samples: 50 | Screening: Drop attention failures, speeders, straight liners | Run the analysis | Make an export sheet

Done.

Long runs are split automatically, so no single step runs longer than the seconds per button press set in Twin Lab. Blindfolding and PLSpredict run once at the start; the bootstrap continues in batches.

Construct reliability and validity | Discriminant validity | Collinearity | Outer model | Path coefficients | Indirect and total effects | R², f², Q²

PLSpredict | Model fit | Export

Export

34 rows and 52 columns written to the **Export** tab of your Google Sheet. From there use File, Download, Comma separated values, and open it in SmartPLS, R, or the Python package.

Suggested file name: driver_scores_synthetic_2026-09-27.csv. Every item is on a 1 to 7 scale with reverse coded items already turned around, and the last nine columns are the construct averages.

First rows

```
respondent_id,city,service,female,tenure_months,arm,AMQ1,AMQ2,AMQ3,AMQ4,AMQ5,PAC1,PAC2,PAC3,PAC4,PAC5,PAC6,PAC7,PAC8,PAC9,PAC10,PAC11,PAF1,PAF2,PAF3,PAF4,PAT1,PA
SYN-K6DBWL,Mumbai,cab,0,71,BASE,5,5,5,5,6,4,4,4,4,5,5,4,5,5,4,4,4,4,3,2,2.5,2.5,2.5,3,3,2,2.5,4,5,5,4,5,5,6,5,5,5,5.2,4.273,3.25,2.5,2.667,3.25,4.75,5.333,5
SYN-PFA2GJ,Pune,cab,0,41,CON1,6,5,6,6,6,5,5,4,5,5,4,5,5,5,5,5,5,4,5,5,4,7,3,3,2,3,2.5,1,1,4,3,5,2.5,4,4,4,5,4,4,4,4,6,6,5,5.8,5.091,2.75,1.5,4,3.25,4.25,4,5.5
SYN-Z6DFZ7,Jaipur,cab,0,46,CON1,5,7,5,5,6,5.5,5,5,5,5,5,5,5,5,5,5,5,5,5,4,5.5,4,5.5,1,2,3,2,2.5,1,1,4,3,2,4,4,5,6,6,5,6,6,5,5,5,5.6,5.227,2,1.5,3,4,5.5,5.667,5
SYN-PJDG56,Pune,cab,0,47,CON1,7,6,6,7,7,5.5,5.5,7,7,7,5.5,7,7,7,7,1,1,2,2,4,2.5,2.5,5,3,4,2.5,2.5,6,5,4,5,6,6,6,6,7,6.6,6.591,1.5,3,4,2.5,5,6,6.5
SYN-4K8VAS,Bengaluru,auto,0,43,BASE,5,5,5,5,5,5,5,5,5,5,5,4,5,5,5,5,7,7,5,5,5,5,1,2,2,2,1,1,1,4,4,3,2.5,4,5,5,5,5,5,5,4,7,7,5,5.636,1.75,1,3.667,3.25,5,4.667
SYN-EKA8QG,Mumbai,moto,0,48,REP1,5,5,5,5,4,5,4,4,4,4,2.5,4,4,4,4,4,4,4,4,3,4,2,3,2.5,2.5,2.5,5,5,4,4,4,4,4,5,4,4,3,3,5,7,7,4.8,3.864,3,2.5,4.333,4,4.25,3.667,7
SYN-6Q47UA,Jaipur,cab,0,41,MGT1,7,6,7,7,7,5.5,7,5.5,5.5,5.5,5.5,5.5,5.5,5.5,5.5,7,5.5,5.5,5.5,1,2,1,3,2.5,4,2.5,3,4,3,2.5,2.5,5,6,5,6,7,6,7,5,5,6.8,5.773,1.75,3,3.333,2.5,5.5
SYN-FHVEAP,Mumbai,cab,0,43,REP1,6,6,7,7,6,4,4,4,5.5,4,4,4,4,4,4,4,3,3,4,3,4,2.5,2.5,4,4,5,4,4,2,3,3,2,3,3,2,7,6,6.4,4.136,3.25,3,4.333,4,2.5,2.667,6.5
SYN-EGKVYE,Delhi NCR,cab,0,63,BASE,6,5,6,6,7,4,5.5,4,4,4,5.5,5.5,5.5,5.5,5.5,4,5.5,5.5,1,2,2,1,2.5,4,2.5,3,3,2,4,5.5,4,4,5,4,5.5,4,4,6,6,4.818,1.5,3,2.667,4.75,4.25,4.667
SYN-9EEPAM,Jaipur,cab,1,51,BASE,6,6,6,6,7,7,5.5,5.5,5.5,5.5,5.5,5.5,5.5,4,5.5,5.5,5.5,5.5,5.5,5.5,5.5,3,2,3,2,2.5,1,4,3,2,2.5,4,6,5,6,5,4,4,4,4,5,6.4,5.364,2.5,2.5,2.333,3.25,5.5
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Research Model

The model in one line

Five Fairwork protections act on how drivers see the algorithm, and on satisfaction, which in turn shapes loyalty, algoactivism, and turnover intention. Economic dependence moderates those responses. Every effect is defined within a driver, as the difference between that driver's twin under a version and under Uber's current rules.

Five dials	How the algorithm is seen	How the driver feels	What the driver does	Who reacts differently
Pay, conditions, contracts, management, representation	Exposure, perceived control, fairness, transparency	Job satisfaction	Loyalty, algoactivism, turnover intention	Economic dependence moderates

Open as a Google Doc

Research questions

Framed for Information Systems Research: two questions about the design of platform governance, and one about the instrument that makes them answerable.

RQ1 DESIGN CONFIGURATIONS OF ALGORITHMIC GOVERNANCE

How do configurations of platform governance features, procedural, economic, contractual and representational, jointly shape platform workers' justice perceptions, satisfaction and behavioural responses, and are these features complements or substitutes within the design space?

Hypotheses: H1a, H1b, H1c, H1d, H2, H3a, H3b

Information systems contribution: the governed platform is an IT artefact whose rules are design choices. The question is about the configuration, not the single feature, which is where design knowledge lives.

RQ2 CONDITIONAL USER RESPONSES UNDER DEPENDENCE

How does economic dependence on the platform condition those responses, specifically when does dissatisfaction translate into within platform resistance rather than exit, and can a representational feature redirect it?

Hypotheses: H4a, H4b

Information systems contribution: boundary conditions on user response to system design. It says which workers a governance choice actually reaches, which is what a platform or a regulator needs before acting.

RQ3 VALIDATED SIMULATION AS A RESEARCH INSTRUMENT

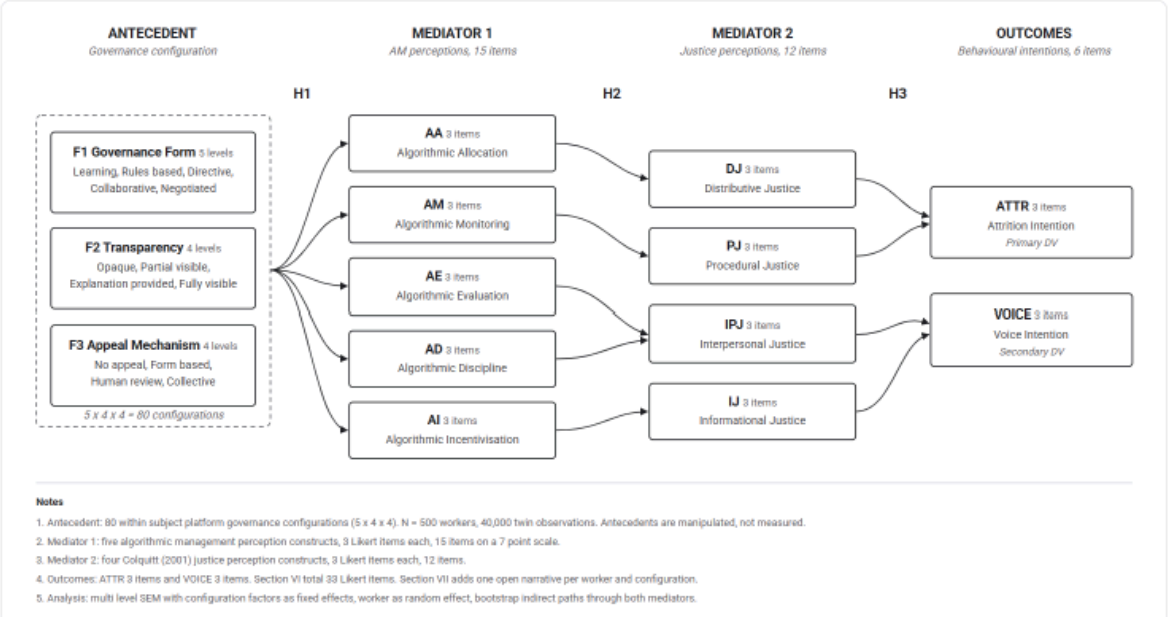
Under what conditions can language model twins, grounded in an individual worker's own survey responses, serve as a valid instrument for counterfactual inference about responses to platform designs that do not yet exist, and what validation protocol establishes those conditions?

Hypotheses: H5, and the validation rules

Information systems contribution: a methodological artefact with stated evaluation criteria, in the design science tradition. It is the instrument that makes RQ1 and RQ2 answerable, and it is reusable beyond this setting.

Research model

Antecedent → Mediator 1 → Mediator 2 → Outcomes. Section VI carries 33 Likert items in total.



Notes

1. Antecedent: 80 within subject platform governance configurations (5 x 4 x 4). N = 500 workers, 40,000 twin observations. Antecedents are manipulated, not measured.
2. Mediator 1: five algorithmic management perception constructs, 3 Likert items each, 15 items on a 7 point scale.
3. Mediator 2: four Colquitt (2001) justice perception constructs, 3 Likert items each, 12 items.
4. Outcomes: ATTR 3 items and VOICE 3 items. Section VI total 33 Likert items. Section VII adds one open narrative per worker and configuration.
5. Analysis: multi level SEM with configuration factors as fixed effects, worker as random effect, bootstrap indirect paths through both mediators.

H1: governance configuration on algorithmic management perceptions. H2: those perceptions on justice perceptions. H3: justice perceptions on behavioural intentions.

Hypothesis	Statement
H1a	Management protections raise perceived fairness.
H1b	Contract protections raise perceived transparency and fairness.
H1c	Pay and conditions protections raise intent to stay.
H1d	Representation lowers algoactivism.
H2	Job satisfaction mediates the links from perceived control, fairness, and transparency to loyalty, algoactivism, and turnover intention; control and fairness also act directly on algoactivism.
H3a	Management and representation are complements: together they reduce algoactivism by more than the sum of their separate effects.
H3b	Pay and conditions are substitutes: together they raise intent to stay by less than the sum of their separate effects.

Gap 2 **Who reacts differently is treated as noise**

Average effects hide that a driver who cannot afford to leave answers differently from one who can. Captive voice is argued theoretically but rarely tested at the level of the individual worker.

QUESTION THAT CLOSES IT

RQ2

For which drivers does economic dependence turn dissatisfaction into resistance rather than exit, and does an institutional voice channel redirect it?

HOW THIS STUDY CLOSES IT

Dependence measured directly, and every version put to every twin, so response surfaces can be compared driver by driver.

Hypothesis	Statement
H4a	Economic dependence weakens the negative link between job satisfaction and turnover intention.
H4b	Economic dependence strengthens the effect of representation in reducing algoactivism.

Gap 3 **Simulation is validated on plausibility, not on counterfactuals**

Fifteen of thirty five generative simulation studies rested on face plausibility. Where validation exists it compares aggregates, not the same person's reaction to a rule they have never lived under.

QUESTION THAT CLOSES IT

RQ3

HOW THIS STUDY CLOSES IT

Held out items, held out behaviour, counterfactual vignettes and conjoint choices, plus eight

Gap 3 **Simulation is validated on plausibility, not on counterfactuals**

Fifteen of thirty five generative simulation studies rested on face plausibility. Where validation exists it compares aggregates, not the same person's reaction to a rule they have never lived under.

QUESTION THAT CLOSES IT

RQ3

Under what conditions can twins built from a driver's own survey answer RQ1 and RQ2, and where do they fail?

HOW THIS STUDY CLOSES IT

Held out items, held out behaviour, counterfactual vignettes and conjoint choices, plus eight adversarial tests, each with a rule fixed before the data are seen.

Hypothesis	Statement
H5	Twin results are interpreted only where they pass the validation rules, lever by lever and outcome by outcome.

Gap 4 **No one asks how good is good enough**

Accuracy is reported without a ceiling. A twin cannot be more consistent than the driver is with themselves, and no study compares twin error with that limit or with what the driver later does.

QUESTION THAT CLOSES IT

RQ3

Under what conditions can twins built from a driver's own survey answer RQ1 and RQ2, and where do they fail?

HOW THIS STUDY CLOSES IT

A two week retest of about 50 drivers gives the ceiling; a three month follow up of about 100 drivers gives a behavioural criterion.

Hypothesis	Statement
H5	Twin results are interpreted only where they pass the validation rules, lever by lever and outcome by outcome.

Research method

Part	What it involves
Design	A survey of drivers with an embedded vignette experiment and a choice based conjoint, then a digital twin of each consenting driver, then the full 243 version design space put to the twins.
Sample	About 500 Uber drivers across five Indian cities and three services, with a target of 33 per city and service cell, recruited at hubs, fuel stations and through driver networks.

Research method

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Design	A survey of drivers with an embedded vignette experiment andl a choice based conjoint, then a digital twin of each consenting driver, then the full 243 version design space put to the twins.
Sample	About 500 Uber drivers across five Indian cities and three services, with a target of 33 per city and service cell, recruited at hubs, fuel stations and through driver networks.
Measures	Nine constructs on 37 items, each with an optional one line reason: exposure, perceived control, fairness, transparency, satisfaction, loyalty, algoactivism, turnover intention, economic dependence.
Experiment inside the survey	Part 10 gives each driver one randomly assigned change and asks 16 follow up questions; Part 10b asks the same driver to rate six versions of Uber drawn from the design space.
Twin construction	One twin per driver, built only from that driver's own answers and written reasons. The instruction is frozen before the main study and every version of it is logged.
Validation before use	Held out items, the five held out behaviour questions, Part 10 and Part 10b, and the eight validity tests. Results are interpreted only for the dials, outcomes and driver groups that pass.
Design space	Five dials at three settings gives 243 versions. Agents test single dials first, then the promising pairs, then the remainder, with every answer written to the sheet.
Analysis	PLS-SEM for the measurement and structural model on the human data; conjoint effects for the dials; within driver differences for interactions; subgroup accuracy for fidelity.
Ethics and openness	Informed consent including a separate statement about building a twin, no personal identifiers in the twin record, the right to withdraw, and pre-registration of hypotheses, the frozen instruction and the validation rules.

Constructs and definitions

Code	Construct	Definition	Where it comes from
EXPO	Algorithmic management exposure	How much of the work is assigned, tracked, rated,, and paid by the app rather than by a person.	Parent-Rocheleau and Parker (2022); Kellogg et al. (2020)
CTRL	Perceived algorithmic control	The driver's sense that the app directs and constrains the working day.	Möhlmann et al. (2021); Wood et al. (2019)
FAIR	Perceived procedural fairness	Whether decisions about ratings, pay, and deactivation are seen as made by a fair process, with a hearing and a way to appeal.	Colquitt (2001); Folger (1977)
TRAN	Perceived transparency	How far the driver understands the rules by which the app decides, and can find and read them.	Rahman (2021); Bucher et al. (2021)
SATI	Job satisfaction	Overall contentment with the work as it is done through the platform.	Cammann et al. (1983), Michigan Organizational Assessment Questionnaire
LOYA	Loyalty	Willingness to stay with the platform and to speak well of it.	Hirschman (1970); Rusbult and Farrell (1983)
VOIC	Algoactivism, or voice	Speaking up, complaining, organising, or working around the app to change how it treats drivers.	Kellogg et al. (2020); Bucher et al. (2021)
EXIT	Turnover intention	Intention to stop driving for the platform.	Rusbult and Farrell (1983); Hirschman (1970)
EDEP	Economic dependence	How far the household depends on platform income, and how costly leaving would be.	Schor et al. (2020); Wood et al. (2019)

Hypotheses

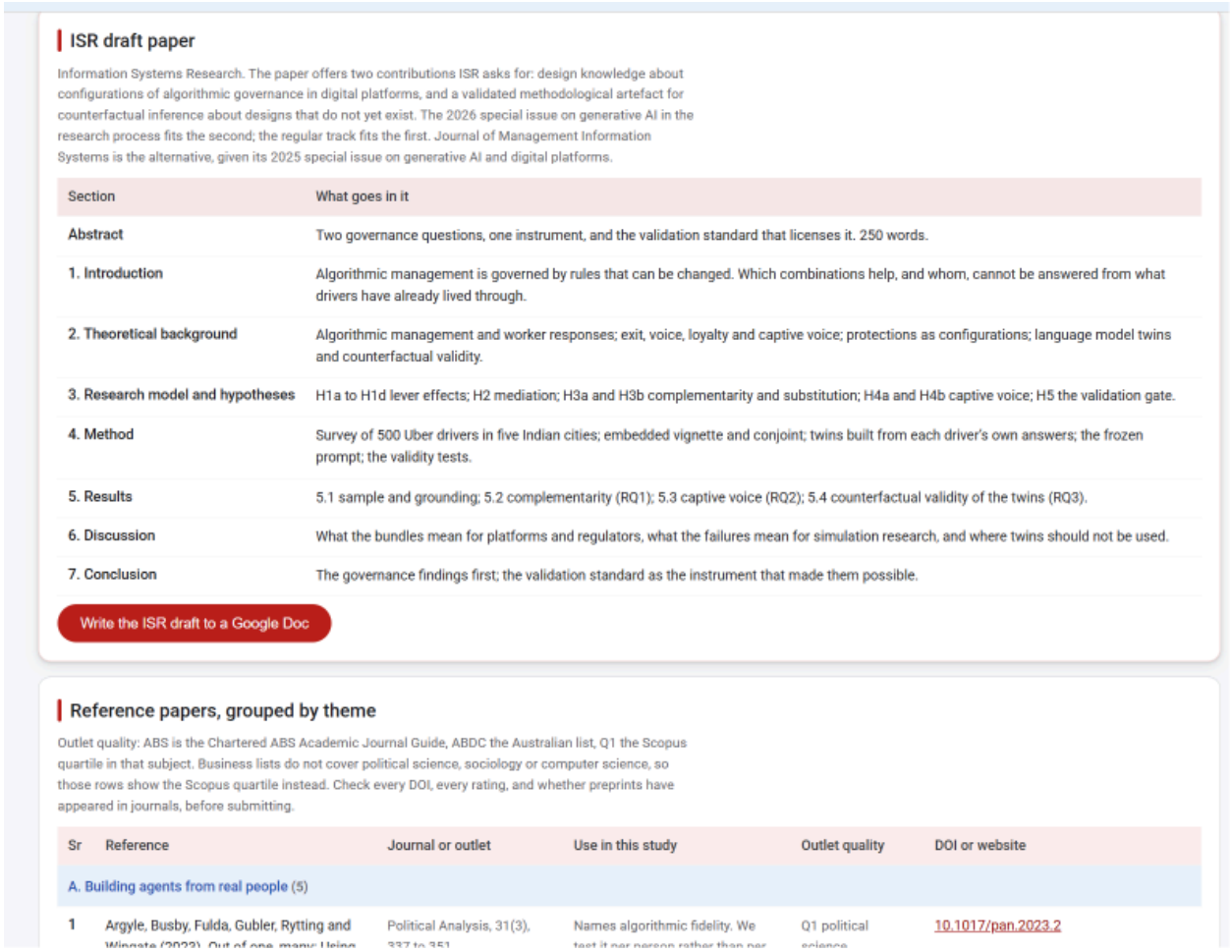
Number	Statement	Citations
H1a	Management protections raise perceived fairness.	Colquitt (2001); Jabagi et al. (2025)
H1b	Contract protections raise perceived transparency and fairness.	Rahman (2021); Höddinghaus et al. (2021)
H1c	Pay and conditions protections raise intent to stay.	Rusbult and Farrell (1983); Schor et al. (2020)
H1d	Representation lowers algoactivism.	Hirschman (1970); Kellogg et al. (2020)
H2	Job satisfaction mediates the links from perceived control, fairness, and transparency to loyalty, algoactivism, and turnover	Farrell (1983); Jabagi et al. (2025);

H3a	Management and representation are complements: together they reduce algoactivism by more than the sum of their separate effects.	Milgrom and Roberts (1995); MacDuffie (1995)
H3b	Pay and conditions are substitutes: together they raise intent to stay by less than the sum of their separate effects.	Milgrom and Roberts (1995)
H4a	Economic dependence weakens the negative link between job satisfaction and turnover intention.	Hirschman (1970); Schor et al. (2020)
H4b	Economic dependence strengthens the effect of representation in reducing algoactivism.	Bucher et al. (2021); Wood et al. (2019)
H5	Twin results are interpreted only where they pass the validation rules, lever by lever and outcome by outcome.	Park et al. (2024); Argyle et al. (2023); Bisbee et al. (2024)

How the twins are validated

Every technique, what it answers, where it runs in this software, and the paper behind it. Techniques marked as fieldwork happen with real drivers rather than in the app.

Technique	What it answers	Where it runs	Paper behind it
Held out survey items	Does the twin hold this person, or a generic driver?	Twin Lab, quality test: 16 of 37 items hidden	Park et al. (2024)
Held out behaviour and intentions (Part 9b)	Can the twin answer five questions never put into its memory?	Twin Lab, Validation tab	Our own extension of the held out idea
Counterfactual validation on Part 10 and 10b	Can the twin predict a reaction to a rule the driver has never lived under?	Twin Lab, Validation tab	Park et al. (2024); Argyle et al. (2023)
Wrong person control	Is a twin about its own driver, or about drivers in general?	Validity tests tab	Scientific Reports (2026) mismatched persona test
Beating simple guesses	Do twins beat no change, the group average shift, and a regression?	Validity tests tab	Scientific Reports (2026); Park et al. (2024)
Placebo change	Do twins welcome any change put to them?	Validity tests tab, needs a live model	Brucks and Toubia (2025) on agreeableness
Impossible version	Do twins notice a contradiction, or rate anything positively?	Validity tests tab, needs a live model	Brucks and Toubia (2025)
Order and wording swap	Is the answer about the change, or about the phrasing?	Validity tests tab	Lyman et al. (2025); Dominguez-Olmedo et al. (2024)
Leave one block out	Does the twin actually use its memory?	Validity tests tab	Park et al. (2024) source comparison
Second AI model	Is the finding about the drivers or about one model?	Validity tests tab, needs a live model	Vaccination fidelity study (2026); Bisbee et al. (2024)
Temperature sweep	How much of the answer is chance?	Validity tests tab, needs a live model	Lyman et al. (2025)
Dispersion check	Do twins vary as much as real drivers, or less?	Statistics, and the Python package	Bisbee et al. (2024)
Accuracy by driver group	Are twins worse for some groups of drivers?	Python package, twin_accuracy	Wang et al. (2025)
Retest ceiling, planned fieldwork	How good is good? Twin error against the driver's own inconsistency two weeks later.	Fieldwork: re-survey about 50 drivers	Park et al. (2024) normalized accuracy
Behavioural follow up, planned fieldwork	Did drivers do what their twins predicted?	Fieldwork: follow up about 100 drivers after three months	Our own extension; no direct precedent found



ISR draft paper

Information Systems Research. The paper offers two contributions ISR asks for: design knowledge about configurations of algorithmic governance in digital platforms, and a validated methodological artefact for counterfactual inference about designs that do not yet exist. The 2026 special issue on generative AI in the research process fits the second; the regular track fits the first. Journal of Management Information Systems is the alternative, given its 2025 special issue on generative AI and digital platforms.

Section	What goes in it
Abstract	Two governance questions, one instrument, and the validation standard that licenses it. 250 words.
1. Introduction	Algorithmic management is governed by rules that can be changed. Which combinations help, and whom, cannot be answered from what drivers have already lived through.
2. Theoretical background	Algorithmic management and worker responses; exit, voice, loyalty and captive voice; protections as configurations; language model twins and counterfactual validity.
3. Research model and hypotheses	H1a to H1d lever effects; H2 mediation; H3a and H3b complementarity and substitution; H4a and H4b captive voice; H5 the validation gate.
4. Method	Survey of 500 Uber drivers in five Indian cities; embedded vignette and conjoint; twins built from each driver's own answers; the frozen prompt; the validity tests.
5. Results	5.1 sample and grounding; 5.2 complementarity (RQ1); 5.3 captive voice (RQ2); 5.4 counterfactual validity of the twins (RQ3).
6. Discussion	What the bundles mean for platforms and regulators, what the failures mean for simulation research, and where twins should not be used.
7. Conclusion	The governance findings first; the validation standard as the instrument that made them possible.

Write the ISR draft to a Google Doc

Reference papers, grouped by theme

Outlet quality: ABS is the Chartered ABS Academic Journal Guide, ABDC the Australian list, Q1 the Scopus quartile in that subject. Business lists do not cover political science, sociology or computer science, so those rows show the Scopus quartile instead. Check every DOI, every rating, and whether preprints have appeared in journals, before submitting.

Sr	Reference	Journal or outlet	Use in this study	Outlet quality	DOI or website
A. Building agents from real people (5)					
1	Argyle, Busby, Fulda, Gubler, Rytting and	Political Analysis, 31(3),	Names algorithmic fidelity. We	Q1 political	10.1017/pan.2023.2

I will give all the reference list later, but the space should be there
And the google sheet tab should be deleted our database is supabase